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HYDROMAGNETIC DISK WINDS IN YOUNG STELLAR OBJECTS AND ACTIVE GALACTIC NUCLEI

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ABSTRACT

We present a general theory for the origin and collimation of centrifugally driven, hydromagnetic winds from the surfaces of Keplerian accretion disks that surround central objects such as young stellar objects (YSOs) or black holes. The theory is fully two-dimensional and makes no assumption about self-similarity of the flows. These winds efficiently extract disk angular momentum and convert the gravitational binding energy released in the accretion process into the mechanical energy of the wind. We investigate solutions under the very general assumption that the magnetic flux in the disk scales as a power law in disk radius r_o as $\psi_o \propto r_o^{3/2\alpha}$. We find analytical solutions for many of the physical quantities in the outflow, such as the wind speed, mass-loss rate, thrust, density, Alfvén surface, etc. We demonstrate that the self-similar solutions of Blandford & Payne ($\alpha = 2$) are a special case of our theory. We find that large classes of models have the generic property that they self-collimate due to the $\boldsymbol{J} \times \boldsymbol{B}$ pinching force arising from the toroidal magnetic field component in the outflow that dominates far from the surface of the accretion disk. We also find that flows that achieve modest values for fast magnetosonic Mach numbers recollimate toward the axis forming magnetic focal points.

We identify a unique *non–self-similar* solution ($\alpha = 3$) that carries a constant current intensity everywhere (the toroidal field scales as $B_\phi \propto r^{-1}$ for this case) and is free of the divergences of self-similar models. We apply our results to both bipolar outflows in the vicinity of YSOs and black hole/accretion disk models for active galactic nuclei (AGNs). We find that modest magnetic field strengths ($\leq 10^4$ G for AGNs, 10 G for YSOs) in the inner portions of these disks are sufficient to drive flows that match all the known observational constraints. In no case do the required magnetic energy densities exceed that of thermal pressure in these disks. An important new result in our theory is an intrinsic scale length to the disk outflow. The mass accretion rate is fundamentally connected to the wind mass-loss rate from the disk because it is the wind that carries off the disk angular momentum. We typically find that the ratio of the wind massloss rate to the accretion rate, $\dot{M}_w/\dot{M}_a$, is 10^{-1} for both the YSO and AGN models. Our results suggest that the highly collimated outflows seen in regions of star formation and in galactic nuclei share a simple physical mechanism that depends only upon the presence of magnetized gas with angular momentum, and the depth of the central gravitational potential well.

Subject headings: galaxies: nuclei — MHD — stars: pre–main-sequence — stars: mass loss

1. INTRODUCTION

Observations of star-forming regions in nearby molecular clouds have, over the last decade, revealed that most, if not all, young stellar objects (YSOs) are associated with energetic bipolar outflows. It is well established that these flows cannot be radiatively driven (e.g., Lada 1985; Cabrit 1989). This strongly suggests that the drive must be hydromagnetic in nature, relying upon the rotation and magnetization of the central engine. There is abundant evidence that YSOs are also associated with accretion disks, which are apparent even for stars as evolved as the T Tauri stars (e.g., Hartmann & Kenyon 1987; Bertout & Bouvier 1989; Beckwith et al. 1989). Two natural sources of hydromagnetic outflow in a star-forming region are from the underlying YSO (Hartmann & MacGregor 1982; Shu et al. 1988) or its surrounding accretion disk (Pudritz & Norman 1983, 1986; Königl 1989). If winds derive from central stars, then they must spin at near breakup velocities in order for sufficient momentum to be transferred to a centrifugally driven wind, as is demanded by the observations. The observed projected stellar rotation speeds are rarely more than $v \sin i \leq 20$ km s^{-1}, which is only a tenth of the stellar breakup speed ($\simeq 300$ km s^{-1}; see, e.g., Vogel & Kuhi 1981; Bouvier et al. 1986; Hartmann et al. 1986; Bouvier 1990). The bipolar flows that emerge from the surroundings of T Tauri stars also appear to be uncorrelated with the rotation rate of the underlying star (Gomez de Castro & Pudritz 1992).

There is now strong observational evidence that outflows and disks are correlated with each other. Observations of forbidden line emission, believed to arise in wind-driven shocks within 100 AU of the central YSO, indicate that the equivalent widths of the forbidden lines correlate very well with the near-infrared colors (which are thought to measure disk properties) of these systems and not with the properties of the underlying star (Edwards et al. 1987; Gomez de Castro & Pudritz 1992). Moreover, Cabrit & Andre (1991) have found that sources with well-developed bipolar outflows have typically 20 times the millimeter flux of sources without outflows. They interpret this result in terms of a link between massive circumstellar disks and outflows.

An accretion disk around a central YSO is for many reasons a simpler drive than a spinning central star. A Keplerian disk locally always rotates near breakup velocity, so that large cen-

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trifugal forces can be tapped to provide an effective centrifuge and hence outflow from the disk. Thus, a wind from an accretion disk may flow from large sections of its surface, which extends from some inner radius near the YSO or its magnetosphere out to scales $\geq 10^3$ larger. Thus, one expects that the outflows are intrinsically bipolar on all scales down to the stellar surface or stellar magnetosphere. The most important aspect of a hydromagnetic disk wind is that it efficiently carries off angular momentum and gravitational binding energy of material within the accretion disk. Thus, the wind mass-loss rate $\dot{M}_w$ and the accretion rate through the disk, $\dot{M}_a$, are fundamentally related to each other. The extracted angular momentum carried out by the wind promotes the disk accretion flow to ever smaller disk radii and, finally, onto the YSO itself (Pudritz 1985). Angular momentum removal is a crucial feature of this mechanism when one remembers that one of the most critical problems in star formation theory is that the galactic gas from which stars form has an angular momentum per unit mass, l, that is 10^5–10^6 times larger than that of a star. It is therefore natural to suppose that this angular momentum is shed through an accretion disk, and the prime mover in this regard is a disk wind.

Bipolar outflows do not arise only in YSOs of course. Extragalactic radio jets have for some time been thought to be associated with accretion disks around massive black holes in active galactic nuclei (AGNs) (for reviews see, e.g., Rees 1984; Bridle & Perley 1984; Blandford 1990). Among the most comprehensive models developed for these systems are hydromagnetic winds that may originate from a spinning black hole (Blandford & Znajek 1977; Phinney 1983; Camenzind 1986) or a surrounding magnetized thin accretion disk (Blandford & Payne 1982, hereafter BP). It is striking that the formation of a centrally rotating black hole must also overcome an angular momentum problem that is as severe as that for YSOs for the simple reason that such an object must also form from galactic gas and must be left rotating at less than breakup speed. For spinning black holes, 10^5 decades of angular momentum must be stripped (e.g., Blandford 1991).

The obvious similarities between outflows in such very different environments implies that the nature of the central object itself is irrelevant, and that only the depth of the gravitational potential well that it creates could be of importance. Observations show that bipolar outflows arise in systems in which an accretion disk is likely to be present. It is therefore natural to ask whether or not there is a universal physical principle at work here. Any unifying idea could only involve the conversion of gravitational binding energy of gas orbiting within the depths of the central gravitational potential well. For a Keplerian disk, the energy release is greatest in the deepest part of the well, i.e., from the inner portions of the disk. In the simplest theory, even the spin of the central object is unimportant, since the observations show that vigorous outflow occurs in systems with slowly rotating T Tauri stars. In the AGN case, where it is likely that a relativistic jet of material is required to explain important phenomena such as superluminal expansion, one may also not require that the source of this relativistic plasma be a spinning black hole (Henri & Pelletier 1991).

A Keplerian (or a self-gravitating) accretion disk with an accretion rate $\dot{M}_a$ releases a known amount of gravitational binding energy of gas as it slowly spirals inward toward the central object. The classic theory of viscous, Keplerian accretion disks (Shakura & Sunyaev 1973; Lynden-Bell & Pringle 1974) supposes that viscous stress in the disk is responsible. In this case, the gravitational binding energy must all go into the heating of the disk, and the angular momentum is transported radially outward through the disk by turbulent stresses. There is nothing in this viscous disk theory, however, that compels one to suppose that bipolar flows would arise. The presence of a modest magnetic field that threads through an accretion disk gives rise to a completely new possibility. The gravitational binding energy that must be released in the accretion process is converted into the mechanical energy of a disk wind. In addition, the angular momentum that must be transported is very efficiently carried off by the wind through the disk surfaces. The wind mass-loss rate $\dot{M}_w$ need only be a tiny fraction of $\dot{M}_a$ for this to occur. The extraction process cannot be 100% efficient, and this leads to some heating of the gas in the disk. Even when ambipolar diffusion of fields within protostellar disks is taken into account however, hydromagnetic disk winds can be shown to still carry off typically 90% of the gravitational binding energy that is released during disk accretion (Königl 1989). Another source of dissipation arises from the low levels of magnetohydromagnetic (MHD) "turbulence" that necessarily arises in any magnetized system, as has been discussed by Balbus & Hawley (1991).

In this paper, we develop the general theory of axisymmetric (two-dimensional), steady, hydromagnetic disk winds that originate from Keplerian accretion disks (the case of self-gravitating disks having been discussed in Pudritz & Norman 1983, 1986). The BP theory restricted itself to self-similar solutions of the hydromagnetic equations, but we present a general analysis for which a large class of non–self-similar solutions exists. The idea that the wind could extract the angular momentum of the underlying disk has been proposed already (BP; Pudritz 1985; Königl 1989). A new feature of our work is to derive a set of general results from this assumption. Thus, the disk angular momentum equation is a boundary condition for our analysis. We show under very general conditions that our two-dimensional winds accelerate gas and self-collimate due to magnetic pinching forces in the wind itself. A generic feature of the solutions is that flows can in fact recollimate to form magnetic focal points at some finite distance above the disk on the outflow axis. This important feature of disk winds implies that MHD shocks, heating, particle acceleration, and jet formation arise quite naturally in self-collimating hydromagnetic flows. A major result of our work is the derivation of analytical expressions for the physical quantities of the wind such as the Alfvén surface, terminal speed, wind density, mass-loss rate, etc. We find that the solutions are governed by two parameters (λ and ε) that scale the wind terminal speed and magnetic flux, respectively. Self-similar solutions are a special case of our more general analysis, and we demonstrate that we recover the BP theory under these conditions. Another major result of these calculations is the identification of a unique non–self-similar solution whose current is everywhere finite, thereby overcoming one of the traditional difficulties of self-similar solutions.

In the next section we discuss the basic physics of two-dimensional hydromagnetic winds, emphasizing the conservation laws along streamlines (and magnetic field lines) in the flow and the collimation of the flow due to forces acting perpendicular to field lines. In § 3 we discuss the disk angular momentum equation and use it to calculate the mass-loss rate per unit area from a steady state, Keplerian accretion disk. We examine the critical surfaces in the wind in § 4, and wind colli-

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mation in § 5. Analytical solutions for important physical quantities are presented in § 6. We apply our results to understanding jets and disks in YSOs and AGNs in § 7. The general reader may wish to skip §§ 4 and 5 on a first reading.

2. BASIC TWO-DIMENSIONAL MHD WIND EQUATIONS

The equations describing a steady, axisymmetric MHD flow have been written down by many authors (e.g., Chandrasekhar 1956; Mestel 1961; Schatzmann 1962; Weber & Davis 1967; Michel 1969; Okamoto 1975; Chan & Henriksen 1980; Blandford & Payne 1982; Pudritz & Norman 1983; Pudritz 1985; Sakurai 1985, Lovelace, Wang, & Sulkanen 1987; Königl 1989; Camenzind 1990; Heyvaerts & Norman 1989). A well-known feature of these ideal, axisymmetric, time-independent MHD equations is that the gas flows along magnetic surfaces which are surfaces of constant magnetic flux. There are a number of quantities such as the energy density, the mass flux, and angular momentum that are conserved on each magnetic surface. Moreover, the simplest ideal MHD equations, namely, Faraday's law for the induction of magnetic fields in moving fluids, and the conservation of mass and magnetic flux show that simple relations exist between the velocity and magnetic fields under the stated conditions.

Rotating, magnetized outflows develop strong toroidal magnetic fields that ultimately are important for collimating the flow. It is therefore convenient to decompose the magnetic field $\boldsymbol{B}$ and velocity field $\boldsymbol{u}$ at any point in a flow into poloidal and toroidal components. In cylindrical coordinates (r, ϕ, z),

$$\boldsymbol{u} = \boldsymbol{u}_p + \Omega r \hat{\boldsymbol{\phi}} , \quad \boldsymbol{B} = \boldsymbol{B}_p + B_\phi \hat{\boldsymbol{\phi}} ,$$

where $\boldsymbol{u}_p$, $\boldsymbol{B}_p$ are the poloidal velocity and magnetic field components, $u_\phi = \Omega r$ is the rotation speed of the flowing gas, and B_ϕ is the toroidal magnetic field component.

The technical difficulty in solving a two-dimensional MHD flow is that forces are transmitted from point to point in directions normal to any given magnetic surface. Thus, in addition to poloidal motion along magnetic flux surfaces, the self-consistent analysis of two-dimensional MHD flow requires that force balance transverse to magnetic surfaces be maintained. The nature of these forces is easy to understand: the centrifugal force of an accelerating particle and the magnetic pressure from both the poloidal and toroidal fields which tend to decollimate the flow must be balanced by the ever-growing pinching force $\boldsymbol{J}_p \times \boldsymbol{B}_\phi$ in the outflowing gas. Under very general conditions, self-collimation of MHD flows can occur. The force balance perpendicular to magnetic flux surfaces gives rise to a generalized version if the plasma confinement condition, the so-called Grad-Shafranov equation (see § 2.2).

For future reference, we note that the equation of motion for a magnetized, warm, gas in a gravitational field is

$$\nabla\left[\frac{1}{2}(u_p^2 + \Omega^2 r^2) + \Phi + h\right] = \frac{1}{4\pi\rho}(\nabla \times \boldsymbol{B}) \times \boldsymbol{B} - (\nabla \times \boldsymbol{u}) \times \boldsymbol{u} , \quad (2.1)$$

where Φ is the gravitational potential, h is the enthalpy per unit mass of the fluid and ρ its density, and the sound speed of the gas is given by $c_s^2 = \rho\, \partial h/\partial \rho$. These equations pertain to both the outflow and the disk. In the latter case a viscous torque may also be present (e.g., Pringle 1981), and we discuss this further in § 3.

2.1. *Flow along Magnetic Flux Surfaces: Conservation Laws*

One of the simplest consequences of axisymmetry is that the poloidal part of the magnetic field [$\boldsymbol{B}_p = (B_r, 0, B_z)$ in cylindrical coordinates] is derivable from a scalar potential. Thus, the conservation of magnetic flux implies that

$$\boldsymbol{B}_p = \frac{1}{r} \nabla a \times \hat{\boldsymbol{\phi}} . \quad (2.2a)$$

The scalar function $a(r, z)$ labels the magnetic surfaces in the sense that $a(r, z) = \text{const}$ defines a surface of constant magnetic flux. Its exact definition is that $2\pi a$ is the magnetic flux through a circle of radius r that is in a plane perpendicular to the axis and centered at distance z on the axis. It is the ultimate aim of the theory, then, to find the potential $a(r, z)$ at every point in the flow.

The Maxwell equation governing the induction of the magnetic field due to fluid motion (known as the induction equation) reduces to $\nabla \times (\boldsymbol{u} \times \boldsymbol{B}) = 0$ in steady state. One consequence of this is that, under axisymmetric conditions, there cannot be a toroidal electric field, so that $\boldsymbol{u}_p \times \boldsymbol{B}_p = 0$. Thus, the poloidal velocity and magnetic field components are everywhere parallel to one another, $\boldsymbol{u}_p \| \boldsymbol{B}_p$. This together with the conservation of mass and magnetic flux implies the relation

$$\rho \boldsymbol{u}_p = \frac{\kappa(a)}{4\pi} \boldsymbol{B}_p \quad (2.2b)$$

between the poloidal field quantities. The function $\kappa(a)$ is a constant on each magnetic surface. A much more physical way of representing this function is to express it in terms of the wind mass-loss rate per unit magnetic flux. The wind mass flow rate through a circle of radius r (as described above) is

$$\dot{M}_w(r) = \int_0^r \rho u_z 2\pi r' \, dr' .$$

Using equations (2.2a) and (2.2b), this can be written as

$$\kappa(a) = 2 \frac{\partial \dot{M}_w}{\partial a} . \quad (2.3)$$

Relations between the toroidal fields are deduced from the induction equation and the angular momentum equation for the flow. Thus, the remaining information in the induction equation is that

$$B_\phi \boldsymbol{u}_p = (\Omega - \Omega_*) r \boldsymbol{B}_p , \quad (2.4)$$

where $\Omega_*(a)$ is a constant on each magnetic surface. Physically it is the angular velocity of the magnetic surface about the axis. It is nearly the angular velocity of the gas in the accretion disk (or star) through which the field lines thread.

Two more conservation laws can be derived from the equation of motion. The ϕ component of equation (2.1) is the conservation of the total angular momentum per unit mass of the flow, and gives

$$l(a) = \Omega r^2 - \frac{r B_\phi}{\kappa} = \text{const} \equiv \Omega_* r_A(a)^2 . \quad (2.5)$$

The specific angular momentum l is a constant on each magnetic surface and has contributions from both the flowing, rotating gas and the twisted magnetic field. The quantity r_A turns out to be the so-called Alfvén radius of the flow. This relation gives considerable insight into the flow dynamics, in

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that, as the material angular momentum of the gas increases due to corotation with the rotor, the toroidal field must steadily increase in order that the total angular momentum transported along a given flux surface remain constant.

The toroidal velocity and magnetic field components can be found from equations (2.4) and (2.5),

$$\Omega = \Omega_*(1 - g)\,, \tag{2.6}$$

$$B_\phi = -\kappa\Omega_* rg(\rho/\rho_A)\,, \tag{2.7}$$

where the function g is

$$g \equiv \frac{r_A^2 - r^2}{r^2}\frac{\rho_A}{\rho - \rho_A} \tag{2.8}$$

and $\rho_A(a)$ is the gas density at the Alfvén point. The Alfvén point is actually a ring on each magnetic surface and is defined as the location where the poloidal speed reaches the Alfvén speed $B_{p,A}/(4\pi\rho_A)^{1/2}$. The collection of Alfvén rings on all of the magnetic surfaces forms the Alfvén surface. In order that physical quantities remain finite at the Alfvén surface, then $r = r_A(a)$ there. The function g measures the angular velocity of the gas with respect to that of the magnetic surface on which it moves. Note that $0 < g < 1$, and g increases monotonically from the origin of the flow to infinity. The solutions (2.6) and (2.7) give the ratio of the rotational to the toroidal magnetic field energy density in the flow in terms of the function g; $4\pi\rho\Omega^2 r^2/B_\phi^2 = (1 - g)^2/g^2$, which clearly shows that the angular velocity of the accelerating gas is steadily reduced while the toroidal field increases. This has an important consequence for the collimation of MHD flows, as will be seen later.

2.2. *Basic Magnetic Geometry and Drive Mechanism*

We have already learned enough from these simple ideal MHD results to understand how the centrifugally driven wind works (see Pudritz, Pelletier, & Gomez de Castro 1991 for a review). Figure 1 presents a cartoon of a field line as it penetrates through the disk at a disk radius r_o which we can call its footpoint. Note that within the accretion disk, the magnetic pressure must never exceed the disk gas pressure, since otherwise instability due to magnetic buoyancy would rapidly set in. Thus, since $B^2_{\rm disk}/8\pi p_{\rm disk} \leq 1$, both the disk field and gas pressure provide the hydrostatic balance condition against the gravity of the central object. The disk cannot self-consistently be pictured to be infinitely thin. Its pressure scale height is controlled by both the gas pressure and magnetic field that thread it. Thus, whatever the external magnetic geometry may be, the field will bend significantly as it enters the disk.

To see how much the field bends within the disk, consider the symmetry of the magnetic field in our disk wind problem. Because protostellar disks are likely formed by collapse along field lines, B_z is an even function of z (same sign above and below disk). The B_r and B_ϕ field components must therefore have odd symmetry because of axisymmetry. Therefore, the radial and toroidal field components must vanish at the midplane. Thus, the magnetic field line cuts the midplane of the disk vertically, as is shown in Figure 1. There is consequently no current (proportional to $\nabla \times \boldsymbol{B}$) at the disk midplane.

Disks in the context of AGNs may also have advected an ordered component of the magnetic field from the molecular gas out of which they formed, so that the same geometry may apply. The reason why molecular cloud magnetic fields are

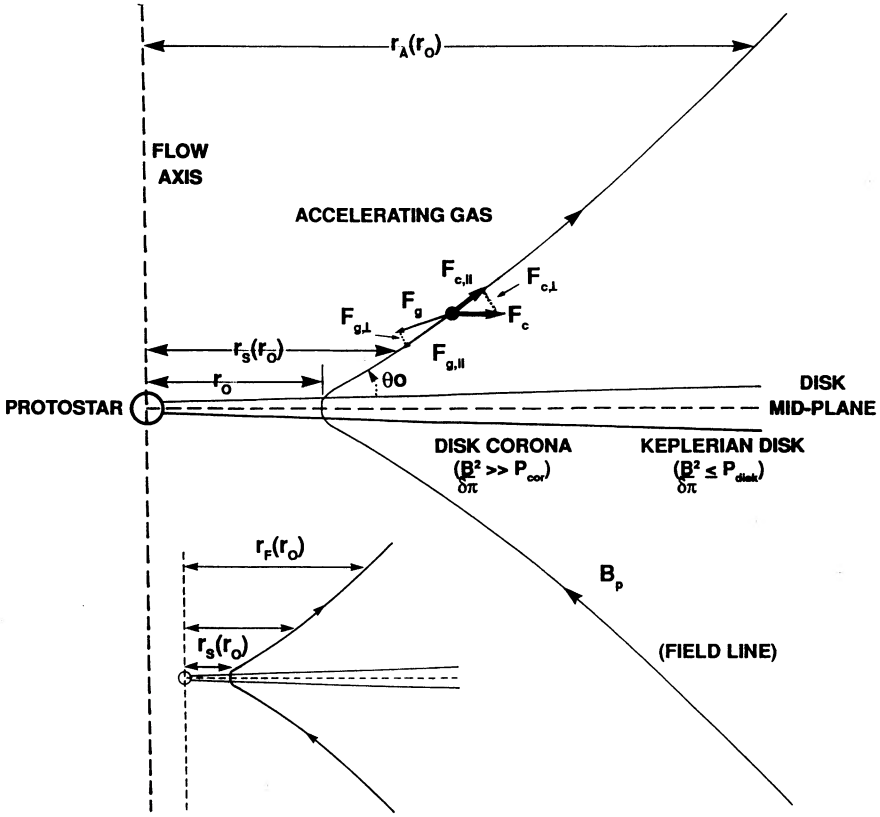


FIG. 1.—Structure of a disk around a YSO (or black hole) with a threading magnetic field and associated hydromagnetic wind. Forces on a particle attached to the field line are shown (see text). The radius r_o is the footpoint of the magnetic field line upon the disk. The magnetic field cuts the disk midplane vertically. The three general types of critical point of flow are also illustrated.

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important progenitors for the fields in accretion disks is that (1) they are well ordered and (2) the magnetic energy density is comparable to the gravitational energy density in molecular clouds. In a thin, Keplerian accretion disk, on the other hand, the magnetic field cannot exceed the gas pressure of the disk, which is tiny compared with the energy in the Keplerian motion. Thus, the problem for star formation regions is not how to generate magnetic field in gas but how to get rid of it (Mestel & Spitzer 1956 invented ambipolar diffusion to accomplish this). If AGN disks are ultimately fed by gas originally in molecular clouds, the same remarks apply.

It is also plausible that AGN disks form in a much more chaotic gaseous environment than a molecular cloud. Ordered fields in such regions would then be quite weak. In this case, one can always appeal to an inertial wave-driven (or turbulent) dynamo to generate strong, ordered fields (Pudritz 1981; Vishniac, Jin, & Diamond 1990; Stepinski & Levy 1990). One caveat here is that closed quadrupolar fields are generated if one assumes that the field exterior to the disk is a vacuum. However, the critical dynamo number for the growth of even versus odd modes is very sensitive to the adopted boundary conditions (e.g., Pudritz 1981). It is likely that if a dynamo operates in a disk that is assumed to have an external corona or flow, then the fastest-growing mode could be that of a dipolar-like field (X.-H. Zhang 1991, private communication). Most of the results on dynamo-generated accretion disk fields have assumed that the disk is surrounded by an exterior vacuum.

Outside the disk, the magnetic field dominates, and at least near the disk surface, one can think of the magnetic field lines as somewhat like rigid wires that control any flow. The opening field geometry shown in the figure is readily provided if one imagines that the disk forms in a magnetized collapse from larger scales. Models of YSO disks employ collapse factors of at least 10. The magnetic field would be highly distorted into a sharply opening, "hourglass" geometry as shown. Such a geometry is ideal for driving the field.

For a sufficiently cold disk atmosphere, the only forces acting on a parcel of gas attached to the field line are the gravity of the central star $\boldsymbol{F}_g$ (if the disk is self-gravitating, this of course also contributes), and the centrifugal force due to the Keplerian rotation of the particle, $\boldsymbol{F}_c$. By decomposing these forces into components parallel and perpendicular to the "wire," one sees that if one goes out along the field line far enough, the effective gravity on the particle vanishes. From this point onward, the gas will be centrifugally accelerated to form a wind. This point, for cold MHD flows, is known as the slow magnetosonic point, labeled $r_s(r_o)$ in the diagram. We have already discussed the significance of the Alfvénic point $r_A(r_o)$. Acceleration continues the length of the lever arm of the torque, which is just the distance out to the Alfvénic point. The final point of interest in the accelerator is the fast magnetosonic point $r_F(r_o)$, at which point all acceleration has ceased and a toroidal field has become so strong that flow collimation starts to occur.

It is important to realize that *thermal pressure is completely irrelvant in the driving mechanism of the wind.* The gas pressure of the corona determines only what the outflow speed from the slow point will be, since the steady state angular momentum equation of the disk determines the disk wind mass-loss rate (see § 3). Claims that hydromagnetic disk winds are "really" thermally driven (e.g., Shu 1991) misconstrue this basic point. While thermal pressure is an important ingredient for the slowly rotating and poorly magnetized Sun, it is insignificant for hydromagnetic flow from rapidly rotating surfaces such as Keplerian disks.

In order for the flow to be well behaved at the Alfvén surface, the apparent singularity in equations (2.4) and (2.5) can be avoided by demanding that

$$\kappa(a) = (4\pi\rho_A)^{1/2} . \tag{2.9}$$

Comparing this expression with that in equation (2.3), we see that the density of the wind at the Alfvén surface is determined by the wind mass-loss rate per unit magnetic flux through the surface. The local Alfvén Mach number m of the flow is then found from equations (2.2) and (2.9):

$$m \equiv \frac{u_p(4\pi\rho)^{1/2}}{B_p} = \left(\frac{\rho_A}{\rho}\right)^{1/2} , \tag{2.10a}$$

so that $m = 1$ at all points on the Alfvén surface. In the strongly super-Alfvénic regime $m \gg 1$ that characterizes our flows, the gas density is significantly smaller than at the Alfvén surface. The fast magnetosonic speed is the speed of a wave in the *total* magnetic field. Accordingly, we define the local fast magnetosonic Mach number n of the flow as

$$n \equiv \frac{u_p(4\pi\rho)^{1/2}}{(B_p^2 + B_\phi^2)^{1/2}} \simeq m(1 - g^2)^{1/2} , \tag{2.10b}$$

where the second expression is valid sufficiently far from the disk that gravity can be neglected in comparison with other forces (e.g., near the Alfvén surface). At all points along the fast magnetosonic (FM) surface, therefore, $n = 1$. We see from equation (2.10b) that while flows can become strongly super-Alfvénic, they need only achieve modest FM numbers. In this regime, the magnetic field is predominantly toroidal field because

$$B_p^2/B_\phi^2 \simeq 1 - g^2 \simeq n^2/m^2 \ll 1 .$$

2.3. *Currents Associated with Disk Winds*

An important consideration bearing upon the stability of MHD winds is how much current they carry. The poloidal current density is, from Maxwell, $\boldsymbol{J}_P = (c/4\pi)\nabla \times \boldsymbol{B}_\phi$. What concerns us is the total current intensity flowing in the z-direction between the axis and a radius r,

$$I_z(r, z) = 2\pi \int_0^r J_z(r', z')r'\,dr' = (c/2)rB_\phi(r, z) . \tag{2.11a}$$

This expression is to be expected from Ampere's law, and, using equation (2.7), it can be evaluated:

$$I_z(r, z) = -\left(\frac{c}{2}\right)\kappa g\Omega_* \frac{(r^2\rho)}{\rho_A} . \tag{2.11b}$$

The crucial point to the overall physics of the outflows is that, in order that the current I_z not diverge at either large or small radii r, we demand

$$B_\phi \propto r^{-1} , \tag{2.11c}$$

so that $I_z = \text{const}$ throughout the flow. Divergences in the current would lead to instabilities in the jet. This, in turn, would lead to a reorganization of field upon the disk until a lower energy configuration is achieved. We note that these divergences come from the assumption that the angular velocity is Keplerian on the disk, and that these could be avoided by introducing an inner and an outer edge to the disk. Nevertheless, with the unique choice given by equation (2.11c) for the

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toroidal field in the wind, it is in fact not possible to arrange simultaneously for self-similar wind, as we will show later (§ 6.8). We will in fact find that the condition (2.11c) leads to a unique and non–self similar solution to the disk wind problem.

2.4. *Flow Energetics: The Bernoulli Equation*

The Bernoulli equation for the conservation of total energy per unit mass of the MHD flow on each magnetic surface can be derived from the equation of motion. Taking the scalar product of $\boldsymbol{B}_p$ with equation (2.1) leads to

$$\tfrac{1}{2}(u_p^2 + \Omega^2 r^2) + \Phi + h + \Omega_*(\Omega_* r_A^2 - \Omega r^2) = E(a) = \text{const} . \tag{2.12a}$$

The last term on the left-hand side of this expression contains the contribution due to magnetic fields. The magnetic field does not explicitly appear in this term, because the toroidal field, generated by differential rotation, has been eliminated by using the angular momentum and induction equations.

The terminal speed of material that flows along a given flux surface can be readily determined from this Bernoulli equation. For gas that has reached large distances from the source, gravity is negligible, and one finds

$$v_\infty \simeq 2^{1/2}\Omega_*(a) r_A(a) .$$

This result, first obtained by Michel (1969), plays an important role in developing the full two-dimensional theory, as we will see later.

Since we are ultimately interested in solving for the magnetic flux function a, we transform the Bernoulli equation above into the form

$$\frac{m^4}{8\pi\rho_A r^2}(\nabla a)^2 + \Phi + h - \frac{1}{2}(1 - g^2)\Omega_*^2 r^2 = \mathscr{E}(a) \equiv E(a) - \Omega_*^2 r_A^2 . \tag{2.12b}$$

It is convenient to reduce the equations to dimensionless form. We first specify a fiducial exterior radius of the outflow, r_e. This may or may not be the outer disk radius (e.g., Pudritz & Norman 1983). The total magnetic flux threading the accretion disk surface divided by 2π is $a(r_e, 0) \equiv a_e$. This allows us to define the dimensionless magnetic flux function ψ as

$$a(r, z) = a_e \psi(r, z) . \tag{2.13}$$

Note that always $\psi \leq 1$. The gas density ρ is made dimensionless with the Alfvénic Mach number m^2 (eq. [2.10]). The angular velocity of a rotating magnetic surface at the radius r_e is Ω_e, so that a dimensionless angular velocity for each surface is

$$\Omega_*(a) = \Omega_e \omega(\psi) . \tag{2.14}$$

At the surface of a Keplerian accretion disk, $\omega = \omega_o \propto r_o^{-3/2}$, so that ψ and ω likely have some simple relation to one another (see § 3).

We scale the density of the gas at the Alfvén surface with respect to quantities evaluated at r_e. We define the dimensionless gas density at the Alfvén surface as

$$\mu(\psi) = \rho_A(a)\frac{4\pi\Omega_e^2 r_e^6}{a_e^2} . \tag{2.15}$$

Finally, the remaining quantities h, Φ, and $\mathscr{E}$ are made dimensionless by dividing them by $\Omega_e^2 r_e^2$ and are denoted with overbars. Employing the above scalings, the Bernoulli equation (2.12) takes the dimensionless form

$$\frac{1}{2} m^4 \frac{r_e^2}{r^2}(\nabla\psi)^2 = \mu\left[\bar{\mathscr{E}} - \bar{h} - \bar{\Phi} + \frac{1}{2}(1 - g^2)\frac{\omega^2 r^2}{r_e^2}\right] . \tag{2.16}$$

2.5. *Flow Collimation: The Grad-Shafranov Equation*

We now must analyze the forces acting orthogonal to the magnetic surfaces. We first rearrange the equation of motion using the Bernoulli constant $\mathscr{E}$ derived above. This is useful because $\mathscr{E}$ can be regarded as a generalized pressure that remains constant on each magnetic surface. The equation of motion then becomes

$$\nabla\mathscr{E} = \frac{d\mathscr{E}}{da}\nabla a = \frac{1}{4\pi\rho}(\nabla \times \boldsymbol{B}) \times \boldsymbol{B} - (\nabla \times \boldsymbol{u}) \times \boldsymbol{u} - \nabla(\Omega_* \Omega r^2) . \tag{2.17}$$

This equation shows that the gradient in effective gas pressure across the two-dimensional flow must be balanced by the right-hand side of the equation, which is the difference between the magnetic tension and the sum of the magnetic pressure from both the poloidal and toroidal magnetic fields, and the outward centrifugal force of the moving gas. This equation therefore describes the confinement or collimation of the flow.

In order to make the link with the theory of magnetic confinement more transparent, we note that equation (2.17) can, after some algebra (see Appendix A for some details), be written in the form

$$\frac{\rho_A - \rho}{\rho} r^2 \operatorname{div}\left(\frac{\nabla a}{r^2}\right) + \frac{1}{2}\rho\nabla a \cdot \nabla\left(\frac{\rho_A}{\rho^2}\right) = 4\pi\rho r^2\left[\frac{d\mathscr{E}}{da} + G(a, g; r)\right] , \tag{2.18}$$

where the functional G contains all the effects due to toroidal velocity and magnetic fields. These include centrifugal terms (decollimating), the magnetic pressure gradient due to the toroidal magnetic field (decollimating), and the hoop stress which pinches the flow [i.e., $(\nabla \times \boldsymbol{B})_p \times \boldsymbol{B}_\phi$]. The two terms on the left-hand side of this generalized Grad-Shafranov (GS) equation involving ρ_A are inertial terms. They are decollimating and correspond physically to gas moving with the poloidal u_p on the curved poloidal magnetic surface. This motion exerts an outward centrifugal force on the surface, thereby leading to decollimation. The remaining term on the left-hand side is a pure magnetic tension term due to the poloidal field component B_p. The magnetic surface can be regarded as an elastic membrane that resists distortion by its tension. Equation (2.18) is a generalization to the case of moving gas of the well-known Grad-Shafranov equation for static plasma confinement (e.g., Shafranov 1965). In the latter static problem, the effective pressure of a hot plasma is confined by the magnetic tension.

Let us now take stock of the number of unknown functions and constraints in the theory. The full set of MHD wind equations contain two unknown functions $a(r, z)$ and $\rho(r, z)$, which are the magnetic flux function and the gas density, respectively. In addition to these, there are four invariants that are functionals of a, namely, ρ_A, r_A, Ω_*, and $\mathscr{E}$. The unknown functions are solutions of the set of two coupled Bernoulli and GS equations, equations (2.16) and (2.18). The functionals Ω_* and $\mathscr{E}$ can be specified once the distribution of magnetic flux and angular

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momentum on the accretion disk is given. This leaves two conditions to be determined. These conditions arise from the requirement that the flow be well behaved as it passes through the two critical surfaces in the wind, namely, the Alfvénic and fast magnetosonic surfaces. We derive these constraints in § 4.

The above prescription when applied to a Keplerian disk leaves only the magnetic flux function on the disk as arbitrary. However, in order to have a steady state accretion disk whose angular momentum is transported away by the wind, this flux distribution must be arranged so that the disk wind is stable. We will in fact find that the requirement that the current in the wind be divergence-free gives a unique value to the flux distribution on the disk.

The dimensionless GS equation is found from equation (2.18), and is

$$\frac{r^4}{m^2}\,\mathrm{div}\left(\frac{m^2-1}{r^2}\,\nabla\psi\right)=\lambda\bar{\eta}\psi\,. \tag{2.19}$$

The function $\lambda(\psi)$ is constant along a magnetic flux surface,

$$\lambda\equiv\mu\,\frac{\omega^2}{\psi^2}\,\frac{r_{\rm A}^6}{r_e^6}\left(\frac{d\ln r_{\rm A}}{d\ln\psi}\right)^2, \tag{2.20}$$

and it plays an important role in controlling the geometry of the flow. We will find that it controls the shape of the Alfvén surface. The quantity $\bar{\eta}$ is a function which in general is of order unity throughout the flow. If we scale the Alfvén Mach number as

$$m^2\equiv\chi\,\frac{r^2}{r_{\rm A}^2}\,, \tag{2.21}$$

then we may write

$$\bar{\eta}=\frac{1}{\chi^2 n_o^2}\left\{2gn_o+\frac{1}{2}\left[(1-g)\,\frac{r^2}{r_{\rm A}^2}+g\right]n_1+\epsilon_2\,, \right. \tag{2.22}$$

$$n_o\equiv\frac{d\ln r_{\rm A}}{d\ln\psi}\,,\quad n_1\equiv\frac{d\ln\mu\omega^2}{d\ln\psi}\,, \tag{2.23}$$

$$\epsilon_2\equiv\frac{r_e^2}{\omega^2 r_{\rm A}^2}\left[(\bar{\mathscr{E}}-\bar{\Phi}-\bar{h})\,\frac{d\ln\mu}{d\ln\psi}+\frac{d\bar{\mathscr{E}}}{d\ln\psi}\right]. \tag{2.24}$$

The quantity ϵ_2 is important only near the accretion disk and is small otherwise. The functions n_o, n_1 are constant along a surface of constant magnetic flux $\psi=$ const. As we will see in § 5, χ is a smoothly varying and nearly constant function in the asymptotic limit of super-Alfvénic flow. Thus, it is clear that $\bar{\eta}$ does not vary significantly, particularly in the super-Alfvénic regime.

2.6. *The Generalized Michel Scaling*

The GS equation we have derived has made no assumption other than steady state and axisymmetry. The important question is, how strongly is λ likely to vary throughout the flow? Here we note that a much more physical way of writing λ gives

$$\lambda=\frac{\Omega_*^2 r_{\rm A}^2}{V_{\rm A}^2}\,c_{\rm A}\,, \tag{2.25}$$

$$c_{\rm A}\equiv|\nabla r_{\rm A}|_{\rm A}^2\,, \tag{2.26}$$

where $V_{\rm A}$ is the particle speed at the Alfvén surface (i.e., the Alfvén velocity), and the gradient of $r_{\rm A}$ is to be evaluated at the Alfvén surface. Since, as we have seen, the asymptotic speed of the outflow is of order $\Omega_* r_{\rm A}$ on any magnetic flux surface $a=$ const, then we physically expect λ to be nearly constant over most of the flow under the most general conditions.

Let us examine how sensitive the position of the Alfvén surface is to the value of λ. Rearranging equation (2.20) with the help of equations (2.23), (2.15), (2.9), and (2.3), we find

$$r_{\rm A}(a)=\lambda^{1/6}\left(\frac{a^2}{\Omega_*\dot{M}_w}\right)^{1/3}\left(2n_o\,\frac{\partial\ln\dot{M}_w}{\partial\ln a}\right)^{-1/3}. \tag{2.27}$$

This formula for the Alfvén point on each flux surface as a function of the magnetic flux, mass-loss rate, and rotation of the rotor is *the generalization to two dimensions of the formula found by Michel (1969) for the Alfvén point in a one-dimensional hydromagnetic flow* (see also Belcher & MacGregor 1976; Pudritz & Norman 1986). In two dimensions there are corrections (the second factor) of $O(1)$ measuring the logarithmic variation of mass loss and Alfvén radius with magnetic flux. Note that the scaling of the Alfvén surface is totally insensitive to λ. This implies that we can accurately calculate the Alfvén surface as long as λ does not vary by more than two decades, which, as we have already suggested, it is unlikely to do.

With this insight as a guide, we will later show that analytical solutions for MHD outflows can be found assuming only that $\lambda=$ const everywhere. This approach is familiar in the context of plasma confinement problems, and we will show that a rich variety of solutions is generated (§ 6). We now turn to the crucial issue of calculating the torque that an outflow can exert on the underlying accretion disk which is the source of the wind.

3. ANGULAR MOMENTUM EXTRACTION FROM ACCRETION DISKS

3.1. *The Disk Angular Momentum Equation*

Within the body of the accretion disk, viscous (Shakura & Sunyaev 1973; Pringle 1981) as well as nonaxisymmetric wave torques (Larson 1990) can transport angular momentum in addition to the magnetic fields. MHD wind torques are distinguished from all others because angular momentum flows primarily vertically out of the disk surfaces rather than radially through it. The angular momentum equation for a steady state, axisymmetric accretion disk threaded by magnetic field and undergoing viscous stress is

$$\mathrm{div}\left(\rho\boldsymbol{u}_p\Omega r^2-\frac{rB_\phi\boldsymbol{B}_p}{4\pi}\right)=\mathrm{div}\,(\tau_{\rm visc}) \tag{3.1}$$

where $\tau_{\rm visc}$ is the viscous torque in the disk. The magnetic stress on the fluid is the second term on the left-hand side of equation (3.1).

The disk angular momentum equation shows that as long as the ordered magnetic energy density in the disk is comparable to thermal pressure, the magnetic wind torque completely dominates the viscous torque (Pudritz 1986). This is demonstrated by modeling the viscous torque as in Shakura & Sunyaev; $\tau_{{\rm visc};r,\phi}\simeq\alpha rp$, where p is the gas pressure in the disk. Then the ratio of the magnetic to the viscous torque is

$$\frac{\text{Wind torque}}{\text{Viscous torque}}=\left(\frac{B_z^2}{4\pi p}\right)\left(\frac{r_{\rm A}}{\alpha h}\right),$$

where h is the scale height of the disk. Since $\alpha\leq 1$, then, the magnetic wind stress easily dominates viscous effects if the magnetic and thermal pressures in the disk are even remotely comparable. The fundamental point here is that viscous

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torques have lever arms that are tiny ($\simeq \alpha h$) compared with those of disk winds ($\simeq r_A \gg h$).

Wind-borne angular momentum loss must therefore predominate if disk fields have energy densities that are comparable to thermal energy density within Keplerian disks. How likely is this? We note that accretion disks are ideal environments for generating magnetic fields starting from weak seed fields. Thus, whether by dynamo action (Pudritz 1981; Vishniac et al. 1990) or through the action of the newly discovered Balbus & Hawley (1991; see also Tagger et al. 1992) instability, magnetic field generation will readily occur in sheared accretion disks. Field amplification will likely saturate at near-equipartition values with the thermal energy density (beyond this, the field bubbles out of the disk due to the Parker instability). The growth rate of the fields locally is several Kepler times, so that they are always very rapidly generated.

We henceforth assume that disks contain sufficient magnetic flux that viscosity is unimportant. Equation (3.1) can then be vertically integrated over z. The result is the angular momentum equation for the disk undergoing wind torque:

$$\frac{1}{r}\frac{d}{dr}(r\Sigma u_r \Omega r^2) = \frac{2rB_z(r, h)B_\phi(r, h)}{4\pi} + \frac{h}{4\pi r}\frac{d}{dr}(r^2 B_r B_\phi)\bigg|_{z=0}, \tag{3.2}$$

where Σ is the disk surface density. The first term on the right-hand side of equation (3.2) is the angular extraction out of the disk surfaces, while the second term is radial transport of angular momentum through the disk by the radial field component.

Adopting the dipolar-like field geometry illustrated in Figure 1 implies that there is no radial angular momentum transport at the midplane. Thus the second term in the transport equation can be ignored. Noting that the accretion rate through the disk is $\dot{M}_a = -2\pi\Sigma u_r r$, the disk angular momentum equation takes the form

$$\dot{M}_a \frac{d(\Omega r_o^2)}{dr_o} + 2\frac{d\dot{M}_w}{dr_o}[2\Omega r_o^2 - l(a)] = 0, \tag{3.3}$$

where equations (2.2)–(2.5) have been used. Note that in the second term $l(a) \gg \Omega r_o^2$, because the Alfvén surface is far from the disk [$r_A(r_o) \gg r_o$].

The disk angular momentum equation (3.3) (see also Pudritz 1985) clearly shows that the disk accretion rate and wind mass-loss rates are linked; i.e., equation (3.3) implies the order-of-magnitude scaling

$$\dot{M}_a \simeq \langle (r_A/r_o)^2 \rangle \dot{M}_w,$$

where $\langle r_A/r_o \rangle$ denotes a suitably chosen fiducial value for the lever arm (in self-similar theory, it is always a constant). Thus, mass accretion and disk wind mass loss are fundamentally linked with each other. Disk winds with long lever arms require very small mass-loss rates in order to drive a much larger accretion rate through the disk. In a Keplerian disk with an accretion rate of $\dot{M}_a$, the product of the wind mass-loss rate and the square of the Alfvén lever arm is fixed by the condition that the specific angular momentum of the accreting matter be carried away.

We must now specify an inner boundary condition for the torque. In the YSO context, we have already noted that T Tauri stars in accretion disks are not observed to rotate very quickly. If the disk continues down to the surface of the star, then it is clear that material in the disk must be substantially reduced from the Keplerian specific angular momentum within a boundary layer (see Pringle 1981). The Keplerian angular velocity (Ω_K) law holds until a radius r_i, inside of which it must fall to the stellar value. For viscous disks, r_i is nearly the stellar radius. There is in this boundary-value view, however, no physics that dictates how a slowly rotating star was achieved in the first place.

A boundary layer need not exist if the central star is strongly magnetized (Bertout, Basri, & Bouvier 1988). A star with a 10^3 G *ordered dipole field* would be able to disrupt the disk at a few stellar radii (Königl 1991; Camenzind 1990). Were this the case, then the total magnetic flux inside the radius r_i would be $\equiv \psi_i$. Numerically it can be estimated as

$$\psi_i \simeq (B_* r_i^2/B_e r_e^2) \ll 1, \tag{3.4}$$

where the later inequality is easily satisfied if the mean stellar field B_* is not much more than 10^3 G.

It is important to realize, however, that *stellar fields are unlikely to be highly ordered.* While strong active regions containing kilogauss fields likely exist on YSOs and other stars, there is no reason to suppose that a large-scale, ordered, dipole component also exists. It is therefore unclear exactly what inner boundary condition should be imposed.

These difficulties are avoided if we demand that the current never diverge as the central flow axis is approached. We will then not have to bother ourselves too much with the ill-understood boundary region, since no divergences will occur in our model. As has already been discussed, the scaling $B_\phi \propto r^{-1}$ neatly provides this.

Mass accretion onto the central star must occur, and the disk wind mechanism is therefore central to the issue of star formation (Pudritz & Norman 1983). If the central star becomes covered with a complicated corona filled with magnetic loops, then the material from the inner disk will accrete onto the star by falling down along magnetic loops.

3.2. *Disk Wind Mass-Loss Rates*

Steady state angular momentum transport strongly constrains the physics of the disk wind. We rewrite equation (3.3) in order to solve for the wind mass-loss rate:

$$d\dot{M}_w = -\frac{\dot{M}_a}{6\omega}\frac{1}{(r_A^2/r_e^2)\omega^{4/3}}d\omega, \tag{3.5}$$

where we assume that the Alfvén surface corresponding to the surface ψ_i is such that $r_A(r_i) \gg r_i$. This result is crucial to keeping the disk wind theory essentially simple. It says that if the Alfvén surface of the outflow is known, then the disk wind mass-loss rate at each radius of the disk is fixed. The details of how the mass loss occurs are irrelevant. The important point is that, since the amount of angular momentum that must be carried away from any disk radius is well known (given that the disk is Keplerian and knowing the accretion rate), then, given a lever arm, the wind mass-loss rate is completely constrained in order to carry off the disk angular momentum. This physics is not unexpected if one recalls the situation in steady state viscous disks. Here the angular momentum must be carried by a combination of the disk viscosity and surface density (e.g., Pringle 1981). Prescribing the disk viscosity fixes the disk surface density in order that steady accretion flow can be maintained. How the disk actually adjusts to that surface density is irrelevant.

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We may use the result (3.5) to solve for the dimensionless gas density at the Alfvén surface (eq. [2.15]):

$$\mu = \varepsilon^2 \left[\frac{1}{(r_A^2/r_e^2)\omega^{4/3}} \frac{d\ln\omega}{d\psi} \right]^2 . \tag{3.6}$$

It is apparent that a second dimensionless parameter arises from this solution of the disk angular momentum equation, namely,

$$\varepsilon \equiv \frac{\dot{M}_a \Omega_e r_e^3}{3a_e^2} \propto \dot{M}_a B_e^{-2} , \tag{3.7}$$

where B_e is a fiducial threading magnetic field strength at the fiducial scaling radius r_e of the disk wind. This parameter is a measure of the total fiducial magnetic flux that threads the disk at r_e. We will later find that it can equivalently be regarded as a measure of the total current flowing through the disk wind. Yet another useful way of thinking about the parameter arises from the Michel scaling for the Alfvén surface; $r_A^3 \simeq a^2/\dot{M}_a\Omega$. Thus the parameter $\varepsilon \simeq \frac{1}{3}(r_e/r_{A,e})^3$ also can be thought of as measuring the ratio of r_e to the Alfvén radius of the field line that passes through r_e. Because of this, we anticipate that $\varepsilon \leq 1.0$. We later show that this range is indeed consistent with disk wind solutions.

Finally, we must consider how the magnetic flux is distributed with radius r_o across the accretion disk. The numerical value of the magnetic flux, $\psi(r_o)$, is constant on the entire magnetic surface that intersects the disk at radius r_o. The physics governing the flux distribution is therefore very important for the details of the outflow structure. The most natural assumption is that it is a power law of disk radius. We therefore write ω as a power law of the magnetic flux function ψ:

$$\omega = \psi^{-\alpha} . \tag{3.8}$$

This model for the disk flux distribution will be used in § 6, where we find self-consistent solutions of the disk wind equations. For a Keplerian disk ($\omega \propto r_o^{-3/2}$), equation (3.8) gives

$$\begin{aligned} \psi &= (r_o/r_e)^{3/2\alpha} , \\ B_o &= B_e[1 + (dr/dz)_o^2]^{1/2}(r_o/r_e)^{(3/2\alpha)-2} , \\ B_e &\equiv (3/2\alpha)(a_e/r_e^2) , \end{aligned} \tag{3.9}$$

where B_o is the total poloidal magnetic field strength on the disk surface, and $(dr/dz)_o = (B_r/B_z)_o$ is the slope of the magnetic field line on the disk surface. The self-similar wind with $\alpha = 2$ is the BP, self-similar scaling for the disk magnetic field, $B_o \propto r_o^{-5/4}$. We later find the value of α that gives the constant vertical current condition discussed in § 2.

We now turn to consider further constraints on the disk wind, namely, its regularity at critical surfaces in the wind.

4. CRITICAL SURFACES IN THE WIND

The Bernoulli and GS equations completely describe the flow from its origin at the disk out to the distant region where collimation occurs. The wind equations contain three well-known critical points where the poloidal velocity achieves the speed of the three types of signal that can propagate in magnetized warm gas. These are the slow magnetosonic, Alfvénic, and fast magnetosonic speeds. In the case of cold MHD flows, i.e., those for which gas pressure in the flow is much less than the magnetic pressure at any point in the flow, the slow point occurs on the rotating source. In our case this is near the surface of the accretion disk. Thus, for the cold MHD flow to accelerate smoothly to infinity, regularity conditions for the equations must be satisfied at the Alfvén and fast magnetosonic surfaces. In this section we demonstrate that it is a generic property of disk winds that they are smoothly and easily accelerated from near the disk surface to far beyond the Alfvénic surface.

The rate of change of the poloidal speed of the gas at any point along a magnetic flux surface is found by taking the derivative of the Bernoulli equation with respect to the poloidal coordinate s. The resulting expression, valid at all points in a flow (pressure and gravity included), is given in Appendix B (eq. [B1]). If it is assumed that the flow is cold and that the Alfvén radius becomes large enough, then gravity and pressure effects are no longer important at the Alfvénic and fast magnetosonic critical surfaces. Equation (B1) can then be written as

$$\begin{aligned} \frac{(n^2-1)(n_s^2-1)}{m^2-1} \bar{c}_s^2 \frac{\partial \ln(m^2)}{\partial s} &= -\frac{\partial \bar{\Phi}}{\partial s} + \frac{1}{2}\frac{\omega^2}{r_e^2}\left[1 - \frac{m^4}{(m^2-1)^2}\left(1 - \frac{r_A^4}{r^4}\right)\right]\frac{\partial r^2}{\partial s} \\ &\quad - \frac{m^4}{2}\frac{\partial}{\partial s}\left(\frac{\bar{V}_{Ap}^2}{m^2}\right) , \end{aligned} \tag{4.1}$$

where n_s is the slow magnetosonic Mach number (eq. [B5]), and where $\bar{V}_{Ap}$ is the dimensionless poloidal Alfvén speed (eq. [B2]). This equation has three critical surfaces. The condition that the flow pass through the Alfvén critical surface $m = 1$ is trivially satisfied. The fast magnetosonic surface $n = 1$ is the second critical surface in the equation and is found at larger physical scales than the Alfvén surface. The third critical surface is the slow magnetosonic (SM) surface $n_s = 1$ and occurs very close to the accretion disk for cold flows. We investigate these three surfaces in the following subsections.

4.1. *Alfvén Surface*

The Alfvén surface is specified by the condition that $m = 1$. The Bernoulli equation (2.16) becomes a quadratic equation in m^2 very near the Alfvén surface. As is well known, this means that the flow has two possible solutions in going through the surface. The regularity condition then specifies how these branches are to be matched. The GS equation is clearly singular at the Alfvén surface, so that the right-hand side of equation (2.19) must vanish for $m = 1$.

We expand the full Bernoulli equation (B1) to first order, and the details are given in Appendix C. The result is

$$\left(1 + \epsilon_1 - \frac{c_A}{\lambda}\right)^{1/2} r_A(\nabla m^2)|_A = \pm 2[\hat{r} - (\nabla r)|_A] , \tag{4.2}$$

where c_A is defined in equation (2.26) and ϵ_1 is a small quantity defined in the Appendix. The vector $(\nabla r)|_A - \hat{r}$ is in fact normal to the Alfvén surface $r = r_A(a)$.

We now consider the regularity condition for the GS equation at the Alfvén surface. Proceeding in exactly the same way as outlined for the Bernoulli equation, one finds that (see Appendix C)

$$(c_2 - c_3\lambda) r_A(\nabla m^2)|_A = 4\lambda(\hat{r} - (\nabla r)|_A) , \quad c_2 = C_2 \cos\theta_A , \tag{4.3}$$

where C_2 and c_3 are positive coefficients defined in the Appendix and $\cos\theta_A$ is the angle between the field line and the Alfvén surface.

The conditions (4.2) and (4.3) are very similar in form. In order that they simultaneously hold, the determinant must vanish. This results in a quadratic equation for c_2, thereby

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constraining the angle of the field line with respect to the Alfvén surface:

$$c_2^2 - (2c_3\lambda)c_2 + [(c_3^2 - 4)\lambda^2 + 4c_A\lambda] = 0 . \quad (4.4)$$

We have ignored the small contributions from $\epsilon_{1,2}$ in writing this (see Appendix C).

The condition that wind solutions exist at the Alfvén surface is that the discriminant of equation (4.4) be positive. This yields the condition that

$$c_A \leq \lambda , \quad (4.5)$$

which implies that $\Omega_* r_A \geq V_A$ (eq. [2.25]). For the Blandford & Payne similarity solution (see § 6.7.) $c_A = \text{const}$. The two roots of equation (4.4) are of opposite sign, and only the positive root corresponds to a flow which becomes super-Alfvénic.

4.2. *FM Surface*

The fast magnetosonic surface is specified by the condition that $n = 1$. In order for the flow to pass smoothly through the fast surface, the right-hand side of equation (4.1) (if pressure and gravity are ignored; otherwise the right-hand side of eq. [B1] is to be used) must vanish when the left-hand side does, while leaving the derivative of m^2 finite.

We confine our attention to the physically most interesting case of flows that are strongly super-Alfvénic, $m^2 \gg 1$. Then the simplest way of deriving the critical condition is to set $n = 1$ in equation (2.10b) and expand in m^2. This leads immediately to the result

$$m_F^2 = \chi_F \frac{r_F^2}{r_A^2} , \quad \chi_F = \frac{3}{2} . \quad (4.6)$$

We use this to find the condition for passing through the FM surface. The asymptotic form for Bernoulli's equation is given by equation (5.1). Combining it with equation (4.6), we find the condition for passing through the FM surface (in the limit of $m_F^2 \gg 1$),

$$c_F \simeq (\tfrac{2}{3})^3 \lambda , \quad (4.7)$$

where the coefficient

$$c_F \equiv \frac{r_F^2}{r_A^2} (\nabla r_A)^2 |_F . \quad (4.8)$$

The result (4.6) automatically guarantees that the right-hand side of equation (4.1) vanishes at the FM surfaces (a nontrivial remark). For completeness we note that in the limit that $m_F^2 = 1 + \delta$ (where δ is a small positive number), $\lambda \simeq c_F$.

4.3. *Flow near the Disk and the SM Surface*

The slow magnetosonic surface is specified by the condition that $n_s = 1$. This surface is the MHD analog of the sonic point in ordinary hydrodynamic wind theory. The surface is located very near to the disk, since the flow is cold and centrifugal force dominates gas pressure. Therefore, the Alfvén Mach number is small ($m \ll 1$) in this regime. BP found that the flow is established within a pressure scale height of the disk surface.

We first determine the opening angle of the magnetic field at the disk surface in order that outflow can occur. For cold MHD flow, BP found that outflow will occur along magnetic field lines that open at angles $\theta \leq 60°$ with respect to the disk surface. This result is easily generalized to include finite pressure effects directly from the Bernoulli equation. Consider a field line characterized by some value of ψ, and let us calculate the velocity of a particle at some point (r, z) very close to the disk surface and compare it with its velocity at the disk surface (r_o, z_o). Since energy is conserved along a field line, we can evaluate Bernoulli's integral (eq. [2.12b]) at these two different points along the surface $\psi = \text{const}$. Subtracting these two expressions, we obtain

$$\tfrac{1}{2}(u_p^2 - u_{p,o}^2) = (h_o - h) - (\Phi_{\text{eff}} - \Phi_{\text{eff},o}^2) , \quad (4.9)$$

where the effective gravitational potential is

$$\Phi_{\text{eff}}(r, z) = -\Phi(r, z) + \tfrac{1}{2}\Omega_*^2 r^2 .$$

Assuming that the two points are only slightly separated by a vector $(\delta r, \delta z)$, then we may expand the above expression to first order. For outflow to occur, we require that $(u_p - u_{p,o}) > 0$. Using the first-order expansion of equation (4.9), we may solve for the minimum opening angle:

$$\tan\theta = \frac{\delta z}{\delta r} < 3^{1/2}\left[1 + \left(\frac{r_o}{\delta r}\right)^2 \frac{(h_o - h)r_o}{2GM}\right]^{1/2} . \quad (4.10)$$

Thus, the opening angle of field lines along which outflow can occur is $\theta < \tan^{-1}(3^{1/2}) = 60°$ if the flow is completely cold (ignore the enthalpy in eq. [4.10]), as BP first showed. Finite pressure, even if small compared with gravity, can assist the flow to get started (the correction term is of order unity). Thus, we see that a critical angle could in fact be relaxed to as much as $\theta < \tan^{-1}(6^{1/2}) \simeq 70°$. Finite pressure effects as well as the physics of disk formation (discussed in § 2) together make it clear that the opening angles of magnetic fields in disk coronae for driving centrifugal winds from the surfaces of accretion disks are easily achievable. These results are in agreement with those of BP, who in fact specifically showed that field lines can be vertical at the disk midplane and that flow would still occur from the disk surface.

The condition of regularity of the outflow at the SM surface is discussed in Appendix D. The regularity of the solution here (the left-hand side of the critical point equation [eq. (4.1)] must vanish at $n_s = 1$) gives rise to an equation which allows the position of the SM point to be evaluated (eq. [D1]).

The Alfvén Mach number at the SM point, m_s, is found from equation (D4) (see Appendix D):

$$m_s^2 = (4\pi\rho_A)^{1/2} \frac{c_s}{B_p|_s} , \quad (4.11)$$

which is the generalization of the BP result (their eq. [4.8]). In the above we must evaluate the poloidal field B_p at the sonic point. However, the SM point is rather close to the disk surface, and we can get a good approximation to m_s by taking the value of the field on the disk surface, $B_p|_s \simeq B_{p,o}$. This approximation underestimates the SM Alfvén Mach number, but we show later (§ 6) that we derive a value very close to that of BP for the self-similar model. Equation (4.11) informs us that indeed $m_s \ll 1$, since $(\rho_A/\rho_o) \ll 1$ and the gas pressure is tiny compared with the magnetic pressure in the disk corona. Thus, the flow, while passing through the SM point, is practically at rest.

Summarizing this section, we have found two conditions, equations (4.4) and (4.8), that must be satisfied by the field in order that the flow pass through both critical surfaces, becoming super–FM after this has been achieved. For any given value of λ, these two conditions are the general two-dimensional constraints required in order to match up numerical solutions smoothly on either side of the critical surfaces.

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5. MAGNETIC SELF-COLLIMATION OF DISK WINDS

In this section we demonstrate that the increasingly toroidal field in the outflow beyond the Alfvénic surface produces a strong $\boldsymbol{J} \times \boldsymbol{B}$ pinching stress that very efficiently collimates the outflow.

The MHD equations clearly show that the flow accelerates at least as far as the Alfvén surface. If the flow continues to widen beyond the Alfvén surfaces, then, since the density continues to decrease, the flow continues to accelerate (mass conservation). In the opposite case, the flow remains nearly at the Alfvén speed. Henceforth we assume that the former case is true, as the observations of bipolar outflows suggest. This limit formally is defined by the condition $m^2 \gg 1$, and for a given magnetic surface, it implies that $r \gg r_A$. The asymptotic form for Bernoulli's equation (eq. [2.16]) is in the limit that $m^2 \gg 1$,

$$m^4(\nabla\psi)^2 = \frac{2\lambda}{(dr_A/d\psi)^2}\left[\left(\frac{r_A^2}{r^2}\right) - \frac{1}{m^2}\right]. \tag{5.1}$$

This can be put into more useful form by using the scaling $m^2 = \chi(r^2/r_A^2)$ (eq. [2.21]) to find that

$$(r\nabla \ln \psi)^2 = \frac{2\lambda}{n_o^2}\frac{\chi - 1}{\chi^3}. \tag{5.2}$$

We note that by using equation (2.11) the FM Mach number n can be written in terms of χ:

$$\chi = \frac{n^2}{2} + 1. \tag{5.3}$$

Equation (5.2) provides a useful insight into the variation of the flux function ψ across the flow because the right-hand side of the equation has a maximum at $\chi = 3/2$. Therefore, $(r\nabla \ln \psi)^2$ takes its maximum value of $(\frac{2}{3})^3\lambda/n_o^2$ at the FM surface.

The GS equation takes on a simplified form in the limit of super-Alfvénic flows, namely,

$$r^2\Delta\psi + r^2(\nabla \ln \chi)\cdot\nabla\psi = \eta\lambda\psi, \tag{5.4}$$

where the function $\bar{\eta}$ reduces to

$$\bar{\eta} \to \eta(\chi) = \frac{4\chi n_o + [\chi - (1/2)]n_1}{\chi^3 n_o^2}. \tag{5.5}$$

We see that if χ is a smoothly varying function, then *the GS equation is essentially linear in this asymptotic regime.* This result that the asymptotic behavior of MHD flows is essentially a linear or quasi-linear problem also may be found in the solutions of BP. Their results for $n \geq 1$ super-Alfvénic flows are derived from equations which become nearly linear Bessel-type equations as well (see, e.g., eqs. [2.25] and [2.26] in their paper). We therefore have a means to develop wind solutions that can be readily used to compare with the large-scale simulations of disk winds now underway (e.g., Stone et al. 1990).

Using self-similar wind theory, BP found that FM flows $n = 1$ collimate parallel to the flow axis. Heyvaerts & Norman (1989) also found cylindrical collimation for a large class of possible flows. For FM Mach numbers larger than some value, $n \geq n_t$, BP found that self-similar flows actually recollimate toward the axis! This raises the possibility of magnetic focal points in outflows after the turning point (where $n = n_t$) has been passed. The importance of such focal points for jet emission has been discussed by Achterberg, Blandford, & Goldreich (1983) and applied by Gomez de Castro & Pudritz (1992). We now demonstrate that this behavior is a general feature of two-dimensional disk winds.

5.1. *The FM Regime* $n = 1$

BP examined the asymptotic regime of their self-similar theory for the case $n = 1$. Their asymptotic solution for m^2 is exactly our equation (4.6) (see eq. [2.29] in their paper). At the FM surface, $\chi = 3/2$ everywhere and is therefore a constant appearing in the GS equation (eq. [5.4]) and the Bernoulli equation (eq. [5.2]). This immediately simplifies the GS equation to

$$r^2\left(\frac{\partial^2\psi}{\partial r^2} + \frac{1}{r}\frac{\partial\psi}{\partial r} + \frac{\partial^2\psi}{\partial z^2}\right) = \lambda\eta\psi. \tag{5.6}$$

Equation (5.6) has the same form as the GS equation for static plasma confinement in which $\lambda\eta$ is a function of ψ. This equation is nonlinear in general, but for our particular power-law model for the magnetic flux threading the disk (eq. [3.8]), $\eta = \text{const}$. Assuming that we look for solutions with $\lambda = \text{const}$, equation (5.6) is a linear problem of familiar form. The general solution for this model is

$$\psi = ar^\nu, \tag{5.7}$$

where $\nu^2 = \eta\lambda$. Note that in order to satisfy both the Bernoulli and the GS equation, only a power-law solution is allowed, and not the more general Bessel function. The solution demonstrates that surfaces of constant magnetic flux correspond to $r = \text{const}$, i.e., *FM flows collimate to cylinders.*

We evaluate η, noting for future reference that, in the disk flux model (3.7),

$$n_o = 2 - \alpha/3, \quad n_1 = 2\alpha - 10, \tag{5.8}$$

as is shown in the subsequent section (see eqs. [6.2] and [6.15]).

In order to satisfy both Bernoulli and GS equations for the $n = 1$ flow, we find that the value of α takes on a unique value, i.e., $\alpha = 6 - [(3/2)/\chi]$. Thus, in order to arrange a very special outflow that remains FM everywhere, the special magnetic flux distribution $\alpha_{FM} = 5$ is implied. This solution corresponds to flux and magnetic field distributions on a Keplerian disk that scale as $\psi \propto r_o^{0.3}$, $B_o \propto r_o^{-1.7}$. There is no reason to suppose that this is a natural flux distribution for the disk to have, however. We will in fact confirm that this is so in § 6. Thus we do not anticipate that outflow ever takes on the special attributes required to maintain FM ($n = 1$) flow everywhere, and we pass on to the important general case.

5.2. *The Super–FM Regime* $(n > 1)$

We have seen that χ is a very smoothly varying function of the coordinates. The fact that the Bernoulli and GS equations (eqs. [5.2] and (5.4) respectively) are nearly linear in the asymptotic limit immediately implies solutions of the form

$$\psi_k \propto (kr)^{\kappa_r/2}J_\nu[\kappa(k)r]e^{kz} \propto [\kappa(k)r]^{\nu+(\kappa_r/2)}e^{kz},$$
$$\nu^2 = \lambda\eta + \left(\frac{\kappa_r^2}{4}\right) \gg 1, \tag{5.9}$$
$$\kappa_r \equiv -r\frac{\partial \ln \chi}{\partial r}, \quad \kappa_z \equiv \frac{\partial \ln \chi}{\partial z}, \quad \kappa^2 \equiv k(k + \kappa_z),$$

where k is real, $\kappa_{r,z}$ are assumed to be very slowly varying functions, and J_ν is a Bessel function. The equations are only

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satisfied by Bessel functions of large order ν (whose asymptotic form is a power law in r). The general solution is a superposition of these modes for both positive and negative values of k

$$\psi(r, z) = \int dk\, C(k)\psi_k(r, z)\,, \tag{5.10}$$

where the coefficients $C(k)$ are to be determined by ensuring that the flow passes through the FM surface.

Inserting the asymptotic solution (5.10) in the Bernoulli equation (eq. [5.2]), we find

$$\left[\nu + \left(\frac{\kappa_r}{2}\right)\right]^2 + r^2 q(z)^2 = \frac{2\lambda}{n_o^2}\frac{(\chi - 1)}{\chi^3}\,, \quad q(z) \equiv \frac{\partial \ln \psi}{\partial z}\,, \tag{5.11}$$

where $q(z)$ is a slowly varying function and can be thought of as a mean value of k.

A turning point with coordinates (r_t, z_t) exists for the magnetic field lines if $q(z_t) = 0$. Flow beyond these points must recollimate toward the flow axis until stopped by the growing centrifugal effects. The turning point can be found analytically in two limits: (1) in the $\lambda/\chi^2 \ll 1$ limit, with $|\eta\lambda| \ll \kappa_r$, and (2) the opposite extreme. In limit (1) we find that χ_t satisfies the cubic

$$\chi_t^3 - \frac{2\lambda}{n_o^2\kappa_r^2}(\chi_t - 1) = 0\,, \tag{5.12}$$

where we note that $\nu \simeq \kappa_r/2 \gg 1$. In the limit of large χ, the solution is

$$\chi_t \equiv \frac{n_t^2}{2} + 1\,, \quad n_t^2 = \frac{2^{3/2}\lambda^{1/2}}{|n_o|\kappa_r} - 3\,. \tag{5.13}$$

Equation (5.13) is one of the most important results of our paper because it shows that recollimation of flow toward narrow focal points is a generic property of MHD disk winds. This focusing condition, when specialized to the self-similar case $\alpha = 2$, has exactly the form found by BP. Since flows that exceed the critical FM number n_t move at speeds exceeding any signal speed in the flow, it is likely that shocks must be a generic feature of flows as they enter these magnetic focal regions. For mathematical completeness the second case is presented in Appendix E.

In general we expect that the recollimating flow cannot reach radii smaller than r_A, since super-Alfvénic flow would come back into Keplerian rotation at r_A. This can be easily seen from the result that $g \simeq (r^2 - r_A^2)/r^2$ for $m \gg 1$ and from relation (2.6).

5.3. *Asymptotic Collimation: Oscillations in Outflow Radius*

At large distances from the source, the only general results known about collimation are due to Heyvaerts & Noman (1989), who find that flows ultimately achieve cylindrical collimation. With this constraint in mind, it is still possible that the flow radius varies in cross section between minimum and maximum values. In fact, the nonlinear, self-similar solutions of Chan & Henriksen (1980) show precisely this behavior.

In order to check on the possibility of oscillations in radius, let us first suppose that the flow did not oscillate in radius. We would then expect that $\kappa_r, q \simeq 0$ in the Bernoulli equation (eq. [5.11]). We show that this condition could only be satisfied for a very specific structure for the disk magnetic field.

First we note from equation (5.3) that for large n, $\chi_\infty \to n^2/2$. Thus cylindrical collimation implies that the Bernoulli equation reduces to $\lambda = \nu^2 n_o^2 \chi_\infty^2/2$. The GS equation informs us that $\nu^2 = \eta\lambda$. We may now eliminate λ between these two expressions and apply the asymptotic form for η (eq. [5.5]). We find that these conditions can only be satisfied if $2(4n_o + n_1) = 1$. Using expressions (5.8), the result implies that in the asymptotic limit $n \gg 1$, the disk field must have a unique flux distribution in order to rise to a flow without radial oscillations at infinity, namely, $\alpha = 15/4$.

Since we are free to specify whatever flux distribution we wish upon the disk, and in any case there is no obvious physical reason for the particular value of 15/4 found above, the theorem of Heyvaerts & Norman must therefore imply that the outflow varies in radius but can still be neatly placed within a bounding cylinder. While this "proof" is not as general as it could be, since it pertains rigorously only to flows in which all the angular momentum is extracted by the disk wind and that have large FM numbers, it nevertheless allows one to understand that oscillations in outflow radius are a generic property of hydromagnetic winds.

Let us now go on to finding analytical solutions for a wide class of disk winds.

6. A CLASS OF ANALYTICAL SOLUTIONS FOR DISK WINDS

The analysis of flow energetics and collimation has shown that there is a function λ which is in general nearly constant everywhere in the wind. We showed that under the most general conditions the Alfvén surface is completely insensitive to variations in the value of λ. Therefore, we will henceforth assume that

$$\lambda = \text{const}\,,$$

which corresponds to a wide, but not exhaustive, class of solutions. We show in the next subsection that then an analytic solution can be found for the Alfvén surface in the wind. From this we will be able to derive analytic expressions for most of the observables in the flow.

6.1. *Alfvén Surface*

Inserting expression (3.6) for μ into the definition of λ (eq. [2.20]), we obtain the differential equation for the Alfvén surface:

$$\frac{\omega'}{\omega^{4/3}}\frac{r_A'}{r_e} = -\frac{\lambda^{1/2}}{\varepsilon} \tag{6.1}$$

where the prime denotes $d/d\psi$. The solution of equation (6.1) is then

$$r_A = r_{A,i}(\psi/\psi_i)^{2-(\alpha/3)}\,, \tag{6.2}$$

where the Alfvén radius for the flux ψ_i is

$$r_{A,i} = \frac{r_e\lambda^{1/2}}{\alpha[2 - (\alpha/3)]\varepsilon}\psi_i^{2-(\alpha/3)}\,. \tag{6.3}$$

Expression (6.2) for the Alfvén surface can also be written as a function of the footpoint of the field line, $r_A(r_o)$, since on a Keplerian disk $\psi(r_o, 0) = (r_o/r_e)^{3/2\alpha}$. Thus equation (6.2) becomes

$$\frac{r_A}{r_o} = \frac{\lambda^{1/2}}{\alpha[2 - (\alpha/3)]\varepsilon}\psi^{2-\alpha} \propto r_o^{(3/\alpha - 3/2)}\,. \tag{6.4}$$

The self-similar BP solution $r_A/r_o = \text{const}$ (i.e., the Alfvén surface is a cone) is recovered for $\alpha = 2$. The physical interpre-

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tation of this result is just that for a field line that has its footpoint at a radius r_o in the disk, the Alfvén point on that field line scales as given by equation (6.4). The gas will flow along the magnetic flux surface $\psi(r_o) = \text{const}$ and is accelerated from the SM point smoothly through $r_A(r_o)$. The formula above shows that *in general the Alfvén lever arm changes depending upon where on the disk the flow originates.* Thus, in general one expects more exotic behavior than suggested by self-similar solutions.

We have thus found an enormous class of interesting non–self-similar solutions that depend upon the flux distribution in the underlying accretion disk. Solutions with $\alpha > 2$ correspond to field lines that converge into a cylindrical sheath of magnetic field and would lead to a "cylindrical sheath" flow. Solutions for $\alpha < 2$ correspond to winds which steadily widen as one moves out from the central axis, somewhat like the delicate unfolding of a many-petaled lotus. We illustrate these two general classes of outflow geometry in Figure 2.

The fact that such a simple procedure allows the vexed issue of finding the Alfvén surface to be so easily resolved deserves further comment. It is made plausible by the realization that, in self-similar flow, a conical shape $r_A \propto r_o$, could be the only self-consistent one, since the terminal speed of the wind must scale with the underlying Kepler speed in the disk. Elaborate mathematical calculation was not needed to deduce this. It appears that the $\lambda = \text{const}$ condition, which arises naturally from the Bernoulli and GS equations, is the more general, simple method of finding this important surface.

The scaling of the Alfvén surface with the control parameter ε,

$$r_A \propto B_e^2 \dot{M}_a^{-1} ,$$

implies that as the magnetic field of the disk is reduced, the Alfvén surface moves toward the disk surface. Thus the lever arm of the wind is reduced. This implies that the wind mass-loss rate in the wind must increase, and the terminal velocity of the wind decrease, in order to carry out the angular momentum. We will demonstrate this below.

6.2. *Terminal Velocities*

We can now solve for the terminal speed of the outflow, which follows from the Bernoulli equation in the asymptotic regime,

$$u_\infty \simeq 2^{1/2}\Omega_* r_A ; \tag{6.5}$$

using equation (6.4), this yields

$$u_\infty = 2^{1/2}\Omega_e r_e \frac{\lambda^{1/2}}{\alpha[2 - (\alpha/3)]\varepsilon} \psi^{2-(4\alpha/3)} \propto r_o^{-[2-(3/\alpha)]} . \tag{6.6}$$

Since the Keplerian velocity decreases outward in disk radius, the only physically reasonable solutions are that u_∞ also decreases with r_o. This implies that $\alpha \geq 3/2$. Thus we have found that physically acceptable models have a range in disk flux magnetic flux distribution

$$\alpha \geq \tfrac{3}{2} .$$

The flow field we have found would be rather complex to disentangle if it could be directly observed. There is a different terminal speed of the flow on each magnetic flux surface $\psi(r_o) = \text{const}$. For a self-similar scaling as an example, a flow originating at the innermost part of the disk r_i has a terminal speed that is a factor of $u_\infty(r_i)/u_\infty(r_o) = (r_o/r_i)^{1/2}$ larger than a flow originating at a radius r_o farther out in the disk. Thus a flow originating at 0.1 AU reaches terminal speeds 10 times larger than those achieved by flows starting at a disk radius of 10 AU. In general, the flow will not be self-similar, so of course these relative values will change. The point is that as one goes down to smaller spatial scales, one should detect ever faster outflow.

The terminal velocity is proportional to the Alfvén radius, so its scaling with ε is the same. Thus, since $u_\infty \propto B_e^2 \dot{M}_a^{-1}$, reducing the magnetic field threading the disk diminishes the terminal speed of the outflow.

6.3. *Wind Mass-Loss Rate*

The differential equation governing the mass-loss rate in the wind in the asymptotic regime is derived by substituting equa-

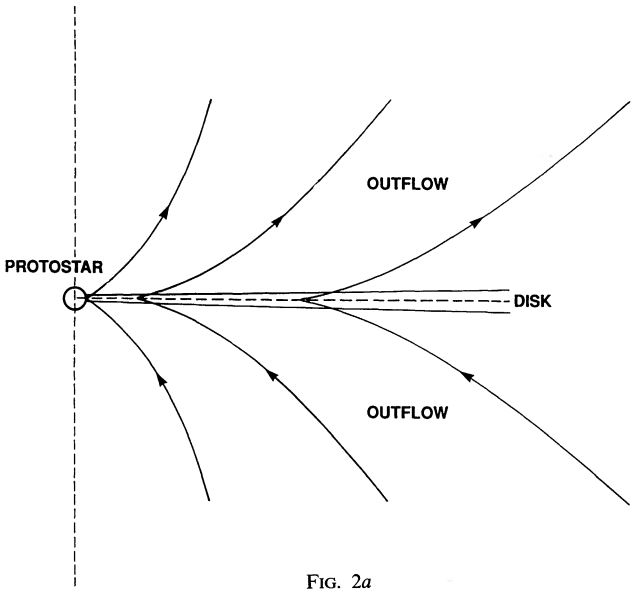


FIG. 2a

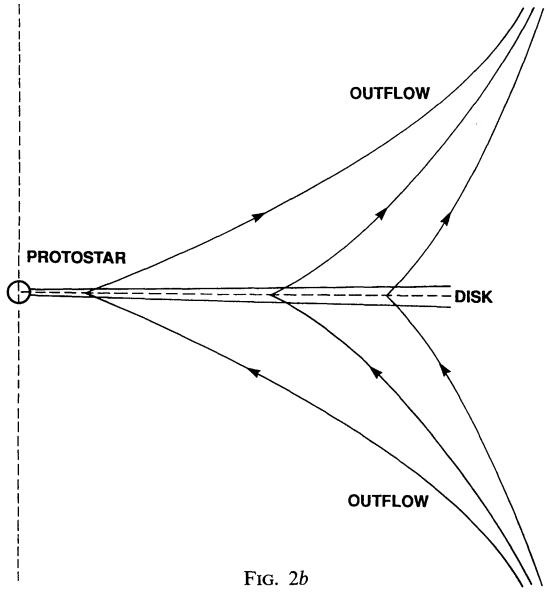


FIG. 2b

FIG. 2.—General non–self-similar theory predicts that there are two general classes of wind geometry, illustrated here. We assume a power-law distribution of magnetic flux upon the disk, $\psi \propto r_o^{3/2\alpha}$ (the BP self-similar model has $\alpha = 2$). (*a*) Generic behavior of the field lines for $\alpha < 2$; (*b*) same as (*a*), but for $\alpha > 2$.

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tion (6.2) in the disk angular momentum equation (3.5):

$$\frac{d\dot{M}_w}{d\psi} = f_m(\alpha)\frac{\varepsilon^2}{\lambda}\dot{M}_a\psi^{2\alpha-5}, \quad f_m(\alpha) \equiv \frac{\alpha^3}{6}\left(2-\frac{\alpha}{3}\right)^2. \quad (6.7)$$

The solutions of equation (6.7) are

$$\dot{M}_w = \frac{f_m(\alpha)}{2\alpha-4}\frac{\varepsilon^2}{\lambda}\dot{M}_a\psi_i^{2\alpha-4}\left[\left(\frac{\psi}{\psi_i}\right)^{2\alpha-4}-1\right]$$

$$\propto \left[\left(\frac{r_o}{r_i}\right)^{3-(6/\alpha)}-1\right] \qquad (\alpha \neq 2), \quad (6.8)$$

$$\dot{M}_w = f_m(2)\frac{\varepsilon^2}{\lambda}\dot{M}_a \ln\left(\frac{\psi}{\psi_i}\right) \qquad (\alpha = 2),$$

where we have assumed that the disk wind mass-loss rate falls to zero at r_i. The $\alpha = 2$ solution is exactly that found by BP. Comparing the scalings of mass loss to mass accretion rates derived above with equation (6.4) for the Alfvénic lever arm, we see that these general solutions confirm our earlier observation that $\dot{M}_a/\dot{M}_w \simeq \langle(r_A/r_o)^2\rangle$. The ratio of the accretion rate to the wind mass-loss rate is indeed large, being

$$(\dot{M}_a/\dot{M}_w) \simeq \varepsilon^2/\lambda \propto \dot{M}_a^2 B_e^{-4}.$$

If the disk magnetic field is reduced, there is a very large increase in the mass-loss rate in the wind. The point is that since a slower wind is driven in weaker disk fields, one must provide much more mass in order to carry off the same amount of disk angular momentum.

6.4. *Wind Luminosity*

The total kinetic power of the wind from each surface of the disk is calculated by integrating over the entire upper and lower surfaces of the accretion disk,

$$L_w \equiv 2 \times \frac{1}{2}\int_{\psi_i}^{\psi} d\psi' \frac{d\dot{M}_w}{d\psi'} u_\infty^2. \quad (6.9)$$

Substituting expressions (6.6) and (6.8) in expression (6.9) and performing the integration, one finds

$$L_w = \frac{1}{2}\frac{G\dot{M}_a M_*}{r_i}\left(1-\frac{r_i}{r_o}\right). \quad (6.10)$$

This is exactly one-half the total gravitational binding energy that can be released in an accretion disk. The remaining $\frac{1}{2}L_{\rm acc}$ is shared between the blackbody radiation and the energy release in the boundary region interior to r_i (see, e.g., Pringle 1981).

An important consequence of this ability of hydromagnetic winds to be highly efficient in energy extraction is that *the mass accretion rate at the outer edge of the disk is greater than the estimate obtained by considering the radiation only.* Only if the torque in the disk is fully viscous do these two rates have to equal one another, after all. There exist also possible flows (Ferreira & Pelletier 1992) in which most of the mass flux is not actually accreted but is diverted in the outflow; in that case, the Eddington limit loses sense.

Real accretion disks do emit radiation. There must be some dissipation of gravitational binding energy into disk heating, and we expect that the coefficient in equation (6.10) will be less than $\frac{1}{2}$. However, detailed calculations of dissipation by ambipolar diffusion in the disk (Königl 1989) show that dissipation only accounts for, at most, about 10% of the losses, the largest share going to the wind. Königl calculates that the ratio of power carried out with the disk wind to the total gravitational power released is $L_{\rm jet}/L_{\rm grav} = 1 - (v_r/v_K)(t_{ni}\Omega_K/2)$ where t_{ni} is the time for a neutral to collide with any ion and v_r is the radial accretion speed in the disk. Since the accretion rate is subsonic, then $v_r/v_K \leq 0.1$ as a conservative estimate for thin accretion disks. Thus, as long as neutrals collide with an ion at least several times during one rotation period in the disk (see detailed calculations in Königl 1989 and Pudritz 1985), then $L_{\rm jet} \geq 0.9L_{\rm grav}$. If the collision rate is less than this, then ambipolar diffusion would become rather severe. In this eventuality, the disk heats up and the ionization fraction rises accordingly.

These results suggest that hydromagnetic disk winds are an excellent mechanical means of transporting away the gravitational binding energy of material in the accretion disk, even in the presence of dissipative mechanisms. The magnetic stress is merely a tool for extracting this energy. Since the mechanical energy of the flow derives from gravitational binding energy of the disk in the first place, it should be no surprise that the scaling (6.10) is obtained. Most of the gravitational binding energy is released in the deepest part of the gravitational potential well, and so most of the outflow's mechanical energy also flows out the central core.

6.5. *Wind Momentum Flux*

This quantity, otherwise known as the wind thrust, is likewise an integrated property of the disk and is

$$F_w \equiv \int_{\psi_i}^{\psi} d\psi' \frac{d\dot{M}_w}{d\psi'} u_\infty. \quad (6.11)$$

The result for $\alpha \neq 3$ is

$$F_w = f_{\rm th}(\alpha)\frac{\varepsilon}{\lambda^{1/2}}\dot{M}_a\Omega_e r_e \psi_i^{(2\alpha/3)-2}\left[1-\left(\frac{\psi}{\psi_i}\right)^{(2\alpha/3)-2}\right]$$

$$\propto \left[1-\left(\frac{r_o}{r_i}\right)^{1-(3/\alpha)}\right], \quad (6.12a)$$

$$f_{\rm th}(\alpha) \equiv \frac{\alpha^2[2-(\alpha/3)]}{2^{1/2}6[1-(\alpha/3)]}$$

while for $\alpha = 3$,

$$F_w = \frac{2^{1/2}}{3}f_m(3)\dot{M}_a\Omega_e r_e \frac{\epsilon}{\lambda^{1/2}}\ln\left(\frac{\psi}{\psi_i}\right) \quad (\alpha = 3). \quad (6.12b)$$

The thrust scales as $F_w \propto \varepsilon \propto B_e^{-2}\dot{M}_a$, so that, as the field threading the disk is reduced, the flux actually increases. The reason for this behavior is just that the mass outflow rate increases enormously as the magnetic field is reduced.

6.6. *Wind Density*

The density can be calculated from equation (2.10). The density at the Alfvén surface is

$$\rho_A = \alpha^6\left[2-\left(\frac{\alpha}{3}\right)\right]^4 \frac{a_e^2}{4\pi\Omega_e^2 r_e^6}\frac{\varepsilon^6}{\lambda^2}\psi^{4\alpha-10} \propto r_o^{6-(15/\alpha)}. \quad (6.13)$$

The density can now be found in the two limiting asymptotic regimes $n = 1$ and $n > 1$. In the former case we use relation (4.6) in addition to equation (6.13) to find

$$\rho r^2 = \frac{\alpha f_m(\alpha)}{2\pi\chi}\frac{\varepsilon^3\dot{M}_a}{\lambda\Omega_e r_e}\psi^{(10\alpha/3)-6} \propto r_o^{5-(9/\alpha)}, \quad (6.14)$$

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where always $\chi = (n^2/2) + 1$ so that, for $n = 1$, we take $\chi = 3/2$. The fiducial density here is quite reasonable, since at any instant it is roughly the total mass of the wind inside the Alfvén surface $r_{A,e}$ divided by the volume $r_{A,e}^3$. For the super–FM branch,we use the appropriate value of n.

The scaling of the wind density with the disk magnetic field is extremely strong, since $\rho \propto \varepsilon^3 \propto B_e^{-6}\dot{M}_a^3$. Thus, decreasing the disk field drastically increases the density of the wind. We expect this, since the mass-loss rate is considerably enhanced in order to carry the disk angular momentum.

6.7. *Angular Momentum in the Outflow*

On each field magnetic flux surface, the total angular momentum per unit mass, $l_{tot} = \Omega_* r_A^2$, is a conserved quantity. The part carried by the field is $l_{mag} = -rB_\phi/\kappa$ (eq. [2.5]). Thus, the fraction of the total angular momentum carried by the toroidal field (the remainder is carried by the particles) in the outflow is $l_{mag}/l_{tot} = (\rho r^2/\rho_A r_A^2)g$.

In the limit that the flow is always *sub-Alfvénic*, the angular momentum is virtually completely carried by the twisted field,

$$l_{mag}/l_{tot} \simeq 1 . \tag{6.15a}$$

In the super-Alfvénic limit appropriate for these outflows, we find that the ratio depends only upon the FM Mach number of the flow.

$$\frac{l_{mag}}{l_{tot}} \simeq \frac{1}{(n^2/2) + 1} , \tag{6.15b}$$

independent of how the magnetic flux is arranged upon the disk. Note that the results (6.15) hold for each field line. We have found, therefore, that the angular momentum in super-Alfvénic flows of modest FM Mach number is carried mainly *by the particles.*

These results are the generalizations of those found by BP (note that in BP the ratio of magnetic to particle angular momenta is calculated).

6.8. *Fixing the Flux Distribution: Currents*

The current intensity can be calculated by substituting for κ, ρr^2, ρ_A, and Ω_* in the equation for the current intensity (eq. [2.11]). It is convenient to evaluate g in either the sub- or super-Alfvénic regimes (the same result is obtained). This gives

$$I_z(r, z) = I_o \psi^{(\alpha/3)-1} ,$$
$$I_o = -\varepsilon \frac{\alpha}{\chi} \frac{c\pi a_e}{2r_e} = -\frac{\alpha}{3\chi} \frac{c\dot{M}_a \Omega_e}{4\pi B_e} \tag{6.16}$$

for flows that become super-Alfvénic. Thus the parameter ε may also be thought of as scaling the total vertical current flowing in the wind and should have values less than unity for self-consistency. Note that if the disk field is reduced, the normalizing current I_o increases.

We see that since ψ must increase in value as one moves outward in disk radius, then if $\alpha < 3$ the current threatens to become singular on the axis. Without an inner disk cutoff, this will happen for self-similar solutions where $\alpha = 2$. If $\alpha > 3$, then without an outer cutoff in disk radius, the current would diverge at infinity. However, for $\alpha = 3$ the current stays finite and constant everywhere. This corresponds to a toroidal field distribution $B_\phi \propto r^{-1}$ as we have discussed in § 2. We have therefore found a unique magnetic field distribution across the disk for which we do not have to introduce cutoffs in disk radius at either large or small scales to avoid singular current intensities. While it is entirely natural to have disks with inner and outer boundaries, both for YSOs and AGNs, this flux law clearly requires the least number of extra assumptions and may therefore be of general importance.

For the particular value $\alpha = 3$, the preceding scalings reduce to

$$\psi \propto r_o^{1/2} , \quad B_o \propto r_o^{-3/2} , \quad r_A \propto r_o^{1/2} , \quad u_\infty \propto r_o^{-1} ,$$
$$\rho \propto (r_o/r)^2 , \quad \dot{M}_w \propto [(r_o/r_i) - 1] , \quad F_w \propto \ln (r_o/r_i) . \tag{6.17}$$

6.9. *Correspondence with the Blandford and Payne Model*

We have seen that self-similar solutions are a particular subset of our general theory. As such, they must correspond to those found by Blandford & Payne (1982). We now show that this is indeed true and, in the process, give a translation of their main parameters into ours. Their control parameters λ_{BP}, κ_{BP} are expressed in terms of our λ, ε:

$$\lambda_{BP} \equiv \frac{l(a)}{\Omega r_o^2} = \left(\frac{3}{8}\right)^2 \frac{\lambda}{\varepsilon^2} , \tag{6.18}$$

$$\kappa_{BP} \equiv k \frac{\Omega r_o}{B_o} = \left(\frac{8}{3}\right)^3 \frac{\varepsilon^3}{\lambda} . \tag{6.19}$$

The "standard model" of BP employs the specific values

$$\lambda_{BP} = 30, \quad \kappa_{BP} = 0.03 ,$$

with an opening angle of the flow at the disk surface $\theta = 32°$.

Using the transformations (6.18) and (6.19), it may be verified that our results in equation (6.8) for the wind mass-loss rate and equation (6.12) for the thrust correspond to the results of BP's equations (5.6) and (5.10), respectively, if the BP parameter measuring the efficiency of angular momentum extraction ($\equiv \epsilon$ in their paper) is equal to $\frac{1}{2}$. The wind luminosity that we find (eq. [6.11]) is identical to the BP result in this case.

BP deduce the condition for super-Alfvénic outflow (formula [3.1] in their paper). Translated into our parameters, their condition becomes

$$\lambda_{min} = \tfrac{1}{2} + (\tfrac{32}{3})\varepsilon^2 \simeq O(1) . \tag{6.20}$$

This is similar to our general constraint for obtaining super-Alfvénic outflow, equation (4.5).

7. APPLICATIONS TO YOUNG STELLAR OBJECTS AND ACTIVE GALACTIC NUCLEI

Our general theory of two-dimensional hydromagnetic disk winds has shown that the gravitational binding energy and angular momentum of Keplerian accretion disks is very efficiently extracted and transported by the wind. We have found an enormous number of possible solutions, among them the BP self-similar model for which conditions are similar throughout the disk. It is likely, however, that during the formation of the disk wind structure, currents carried out by the outflow adjust in such a way as to leave the jet stable and in a minimum-energy state. In this eventuality, we have shown that the outflow toroidal field must scale as $B_\phi \propto 1/r$, and the magnetic flux distribution is uniquely determined ($\alpha = 3$). This scaling produces a flow that is *not self-similar*. Among the new possibilities raised by such a solution is that there is an intrinsic length scale for the Alfvén surface (in fact, all models with $\alpha > 2$ have this property).

In this section, we apply the results of § 6 (with $\alpha = 3$) to the problem of outflows in YSOs and AGNs. Two parameters, λ

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and ε, must be determined for each of these systems, and then one can deduce most physical properties of the resulting outflows. In the YSO context, we will show that the momentum discharge in the disk wind is easily sufficient to drive the CO outflow seen on much larger scales. In the AGN case, this theory readily produces jets with mechanical luminosities of the same order of magnitude as the optical and X-ray luminosity of the nucleus, $\geq 10^{45}$ ergs s^{-1} (assuming a central black hole mass of $\geq 10^8\ M_\odot$).

In both cases, we predict that disk winds drive the jet and wind phenomena observed; that magnetic self-collimation of the flows occurs; and that the outflows will recollimate to magnetic focal points at some finite distance above the disk. MHD shocks will occur at these points, and the radiation produced can readily account for some observational features of spectra in these different systems. In going from YSOs to AGNs, one need only turn up the mass of the central object and adjust the two relevant parameters. We will see, however, that these parameters *take similar values for these two systems.*

In our two-parameter theory, λ does not depend on physical variables such as mass-loss rate, etc. However, the second parameter, ε (defined in eq. [3.7]), depends upon the accretion rate through the disk, the mass of the central object, and the disk magnetic field. The value of the parameter at the inner radius of the disk wind is

$$\varepsilon_i = \varepsilon\left(\frac{r_i}{r_e}\right)^{3/2-3/\alpha} = \varepsilon\left(\frac{r_i}{r_e}\right)^{1/2} \quad (\alpha = 3)$$
$$\varepsilon_i = \frac{\dot{M}_a (GM)^{1/2}}{3B_i^2 r_i^{5/2}}, \tag{7.1}$$

where B_i is the poloidal magnetic field strength at the inner edge of the disk. The opening angle of the magnetic field increases as we go from r_i to r_e, and we scale our flux at r_e so that $a_e = 2[1 + (dr/dz)^2_{o,e}]^{-1/2} B_e r_e^2 \simeq B_e r_e^2$.

We have already argued that the value of ε is less than or equal to unity 1 throughout the wind-producing zone on the disk. In fact, we can define a fiducial scale length (r_e) for the outflow from the disk surface by the condition

$$\varepsilon(r_e) \simeq 1 . \tag{7.2}$$

We will assume in all the subsequent calculations that the disk wind system has achieved a configuration such that $\alpha = 3$.

7.1. *Young Stellar Objects*

The first phase in the creation of a young stellar object is likely to be the formation of a disk that may be quite massive (e.g., Beckwith et al. 1989). Outflow from self-gravitating rotors has been discussed at length (e.g., Pudritz & Norman 1986). While this is of importance at the very earliest stages when star formation starts to proceed by accretion out of the disk (solar mass disks are postulated for sources such as L1551; Pudritz 1988), in the later stages, a massive protostellar core must be built up. The innermost parts of the surrounding somewhat exhausted disk will then be dominated by the central object. Under these circumstances, the disks are Keplerian. The most energetic outflow phenomena in the YSO will be driven from this innermost region, since this is where the gravitational potential well is the deepest.

We first establish a plausible value for ε_i. Detailed studies of T Tauri stars reveal that many may still be accreting gas from a surrounding accretion disk. Accordingly, we adopt our scaling for ε_i by choosing parameters appropriate for T Tauri systems (these have both disks and jets). If one applies viscous disk theory to the observations of T Tauri stars, then (see, e.g., Bertout & Bouvier 1989) one deduces that a fiducial accretion rate of order $\dot{M}_a \simeq 1 \times 10^{-7}\ M_\odot\ \mathrm{yr}^{-1}$ falls onto a solar mass central star. We must emphasize that this rate, in our theory, could be significantly underestimated because of the high efficiency with which the winds carry off gravitational binding energy. Nonetheless, we use this observationally based value.

The radius of a T Tauri star is typically several solar radii, and we adopt $R_* \simeq 3\ R_\odot$ for definiteness. Since these stars likely have a strong, presumably chaotic magnetosphere with a central field of order 10^3 G, the inner disk is truncated by the magnetic stress due to the magnetosphere at a magnetopause radius of order $\simeq 5\ R_*$ (Bertout et al. 1988). With these considerations in mind, we adopt $r_i \simeq 15\ R_\odot = 10^{12}$ cm (note that this is somewhat larger than the fiducial value employed by Königl 1991). The field strength in the disk just outside the magnetopause radius can be no more than in equipartition with gas pressure there, so that we adopt $B_i \simeq 10$ G for the inner disk field. Note that this implies a disk field strength at a disk radius of 100 AU of no more than $B_o(r_o = 100\ \mathrm{AU}) \simeq 0.3 \times 10^{-3}$ G. This is a very reasonable value for the outermost disk field strength, since measurements of the magnetic field in the surrounding larger scale, lower density molecular cloud find values of order 0.3×10^{-4} G (e.g., Myers & Goodman 1988). Thus the disk field is readily explained as having originated on larger scales and having been compressed in the collapse that formed the accretion disk and its threading magnetic field.

With these estimates, we find that for YSOs a fiducial value for one of our two fundamental parameters is

$$\varepsilon_i = 0.24\left[\frac{(\dot{M}_a/10^{-7}\ \mathrm{yr}^{-1})(M/M_\odot)^{1/2}}{(B_i/10\ \mathrm{G})^2 (r_i/10^{12}\ \mathrm{cm})^{5/2}}\right]. \tag{7.3}$$

An immediate consequence of this value is that there is a characteristic radius r_e to the wind zone. Combining equations (7.1)–(7.3), we deduce that

$$r_e \simeq \frac{r_i}{\epsilon_i^2} = 17.4 r_i = 1.74 \times 10^{13}\ \mathrm{cm} . \tag{7.4}$$

Thus, if physical processes drive disk wind structures into a minimum-energy state as we are assuming here, then vigorous outflow derives from the innermost several astronomical units of the Keplerian disk.

We now constrain the second parameter of the theory, λ, by considering the amount of linear momentum carried by the disk wind. A generally accepted view of bipolar CO outflows observed on large scales, 0.01–1 pc, is that they are driven by a fast, highly collimated wind that originates from the central star or disk. If this is true, then the momentum carried by the disk wind is responsible for the observed momentum carried in the CO flows. Observations show that the thrust carried in CO flows is enormous compared with the bolometric luminosity of the central YSO (Lada 1985). A recent reworking of the data shows that (Cabrit 1989)

$$\frac{F_{w,\mathrm{CO}}\, c}{L_{\mathrm{bol}}} \simeq 400 . \tag{7.5}$$

The bolometric luminosity arises from both the young stellar object and the accretion onto it. We assume that the lumi-

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nosity is due to accretion onto the star via flow along the magnetosphere from r_i down to the stellar surface (e.g., Königl 1991), so that

$$L_{\rm bol} = \xi \frac{G\dot{M}_a M_*}{R_*}, \tag{7.6}$$

where ξ is an efficiency parameter measuring the conversion of energy released in the guided infall onto the stellar surface to radiation. Estimates of the parameter vary, but we shall adopt a fiducial normalization of $\xi \simeq 0.2$. The total thrust provided by the disk wind is found by using equations (6.12), (6.17), (7.1), and (7.6) to deduce that

$$\frac{F_w(r_e)c}{L_{\rm bol}} = 1.25 \times 10^3 \left[\frac{1}{\lambda^{1/2}(\xi/0.2)}\right]. \tag{7.7}$$

We see that, indeed, the thrust observed in the CO flow can be explained through the impulse of an underlying disk wind if (equating expressions [7.5] and [7.7])

$$\lambda^{1/2} = \frac{3.17}{\xi/0.2}. \tag{7.8}$$

We have now fixed our two adjustable parameters, and can now calculate the remaining physical quantities using the relations found in § 6. The Alfvén surface of the wind is, from equations (6.4), (6.17), and (7.1),

$$\frac{r_{\rm A}(r_o)}{r_o} = 4.42\left(\frac{r_i}{r_o}\right)^{1/2}. \tag{7.9a}$$

This implies that the Alfvén point for gas flowing from the radius r_i is at

$$r_{{\rm A},i} = 4.42 r_i = 4.42 \times 10^{12} \text{ cm}. \tag{7.9b}$$

The Alfvén radius for flow leaving the radius r_e is $r_{{\rm A},e} = 1.06 r_e = 1.81 \times 10^{13}$ cm. We derive this using the assumption that $\varepsilon \simeq 1$ at r_e, which we see is indeed self-consistent.

The terminal speed in the wind is similarly evaluated:

$$u_\infty = \frac{2^{1/2}\Omega_i r_i}{3} \frac{\lambda^{1/2}}{\varepsilon_i}\left(\frac{r_i}{r_o}\right) = 710\left(\frac{r_i}{r_o}\right) \text{km s}^{-1}. \tag{7.10}$$

Thus the outflow from the innermost part of the disk can achieve speeds as high as $\simeq 700$ km s^{-1}. Given that forbidden line widths for these systems are 200–400 km s^{-1} broad, and that these velocities represent shocked gas that must have slowed down from fast, preshocked gas, this is entirely reasonable. It is important to keep in mind that the outflow velocity predicted in equation (7.10) *scales steeply with the source radius*. Thus, for flow leaving the characteristic radius r_e, $u_\infty(r_e) = 1.49\Omega_e r_e = 40.8$ km s^{-1}. Thus, from a region of the disk extending out to several astronomical units from r_i, terminal speeds drop precipitously. This is important for the production of forbidden line emission at the magnetic focal points in the flow because a variety of line excitation conditions are required to explain the observations (Gomez de Castro & Pudritz 1992).

The total mass-loss rate in the flow, given these parameters, is a tiny fraction of the mass accretion rate (using eqs. [6.8], [6.17], and [7.1]):

$$\dot{M}_w(r_e) = \frac{f_m(3)\varepsilon_i^2}{2\lambda}\left(\frac{r_e}{r_i} - 1\right)\dot{M}_a = 0.21\dot{M}_a, \tag{7.11}$$

or $\dot{M}_w = 2.1 \times 10^{-8}\ M_\odot$ yr^{-1}. These rates agree with the inferred mass-loss rates in the optical jets associated with YSOs (see, e.g., Mundt, Brugel, & Buhrke 1987). The result underscores the fundamental point that the efficiency of a disk wind is very high in removing angular momentum. A mass-loss rate about 10% that of the accretion rate will suffice.

We estimate the wind density in the super-Alfvénic part of the outflow using equations (6.14), (6.17), (7.1), as well as (7.9a):

$$n(r) = \frac{\rho(r)}{\mu m_p} = 5.0 \times 10^7 \frac{1}{\mu\chi}\left(\frac{r_o}{r_i}\right)\left[\frac{r_{\rm A}(r_o)}{r}\right]^2 \text{cm}^{-3}, \tag{7.12}$$

where μ is the reduced mass and one is in the limit that $r \gg r_{\rm A}$. It is interesting to note that some of the forbidden lines observed in YSOs require shocks with densities approaching these values (Gomez de Castro & Pudritz 1992).

Finally, we note that flows will recollimate at a modest FM Mach number. Our basic recollimation FM Mach number n_t was given in equation (5.13):

$$n \geq n_t = (2^{3/2}\lambda^{1/2} - 3)^{1/2} = 2.44, \tag{7.13}$$

where we have taken $k_r \simeq 1$. Thus flows which achieve the modest FM Mach number of 2.4 must recollimate and form magnetic focal points. As we have already noted, the flow may undergo a shock at these points. Material would be reaccelerated and then likely undergo recollimation once again. While we have not yet performed any detailed calculations beyond the first recollimation point, it is highly likely that a knotted, optically visible jet structure will result.

As previously argued, recollimation does not seem to require that unreasonably large currents flow in the jet (see Lovelace et al. 1987) if one bears in mind that the Keplerian law and thus the current profile no longer apply near the axis. Using equation (6.16) for the current flowing in the disk wind from a YSO, we find the modest value

$$I_{o,\rm YSO} = \frac{1}{(n^2/2) + 1}\frac{c\dot{M}_a\Omega_i}{4\pi B_i} \simeq \frac{0.61 \times 10^{13}}{(n^2/2) + 1} \text{ A}.$$

7.2. *Active Galactic Nuclei*

The analysis for AGNs follows along in the same manner as in the previous section. In order to derive values for our two parameters in the AGN context, we make use of the standard theory of accretion disks. This theory is supposed to explain the blackbody, UV spectrum of AGNs provided that the accretion rate is of order $\dot{M}_a = 0.1\ M_\odot$ yr^{-1}. The magnetic field in the innermost portion of the disk can be evaluated in the usual manner by assuming that the field strength is of the order of the equipartition value with the pressure in the disk. Unlike the YSO case, the disk is supported predominantly by radiation pressure on these scales. The disk field is then of order 10^4 G at a distance of order 10 black hole radii. We find that the "standard model" parameters of BP for which

$$\varepsilon_i = 0.34 \tag{7.14}$$

provide a good starting point for our own scaling as well. For this value, the accretion rate given above, a black hole mass of $M_{\rm bh} = 10^8\ M_\odot$, and an assumed inner radius $r_i = 10 r_G$, the field strength required is a modest $B_i = 0.4 \times 10^4$ G. Note that the innermost radius should be larger than $3r_G$, since one no longer has a Keplerian disk inside these scales.

With this value for ε, the outermost radius of the outflow in the $\alpha = 3$ context is

$$r_e = 86.5 r_G. \tag{7.15}$$

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We also adopt the value of λ given in the BP standard model, since it fits the data quite well. Thus

$$\lambda^{1/2} = 5 , \tag{7.16}$$

from which we deduce that the typical velocity of the disk wind is subrelativistic,

$$\left\langle \frac{u_\infty}{c} \right\rangle \equiv \frac{F_w(r_e)}{\dot{M}_w} = 0.38 . \tag{7.17}$$

We note that, although the scales, central mass, and velocities are very different, the underlying AGN disk wind parameters, λ and ε_i, are very close in value to the YSO parameter derived above! As an example, we note that the Alfvén point for gas accelerated from the innermost part of an AGN disk accordingly lies at a distance of

$$r_{A,i} = 4.90 r_i = 49 r_G . \tag{7.18}$$

The total mass-loss rate from the AGN disk is

$$\dot{M}_w(r_e) = 0.09 \dot{M}_a , \tag{7.19}$$

or $\dot{M}_w \simeq 0.9 \times 10^{-2}\ M_\odot\ \mathrm{yr}^{-1}$. The density in the super-Alfvénic part of the outflow is then

$$n(r) = 2 \times 10^8 \frac{1}{\mu\chi} \left(\frac{r_o}{r} \right)^2 \mathrm{cm}^{-3} . \tag{7.20}$$

We note that for a FM Mach number of 4, $\chi = 8.5$.

Finally, the critical FM number for the recollimation of the flow is

$$n_t = 3.3 , \tag{7.21}$$

so that once again flows with modest FM Mach numbers must recollimate and converge toward the axis forming focal points.

The fiducial current carried by an AGN disk wind with the parameters discussed above is

$$I_{o,\mathrm{AGN}} \simeq 1.5 \times 10^{17}\ \mathrm{A} ,$$

which is comparable to the power requirements of electromagnetic jet models studied by Lovelace et al. (1987). Thus, centrifugal disk wind models require no larger currents than do electromagnetic models for maintaining steady outflows.

The preceding estimates are quite reasonable for radio jets. The fact that the characteristic speed is subrelativistic corresponds well with radio observations of hot spots in the sources 3C 20, 3C 33, 3C 111, 3C 123, and Pic A (Meisenheimer et al. 1989). Detailed calculations show that the hot spots in Cyg A are readily understood in the context of subrelativistic flow (Roland, Pelletier, & Muxlow 1988).

Superluminal motions of radio knots in AGNs require the presence of relativistic bulk motion with Lorentz factors $\gamma \simeq 10$. These relativistic motions are observed only on scales smaller than $\simeq 1$ kpc from the central engine. These bulk relativistic motions do not necessarily arise from either the inner disk or a rotating black hole, but in an e^+-e^- *pair creation catastrophe* on the flow axis beginning at scales of order $10 r_G$ (Henri & Pelletier 1991; Pelletier et al. 1991). In this mechanism, the pairs are reaccelerated by Alfvén wave turbulence of low amplitude ($\lesssim 10\%$). Because of this acceleration, they do not annihilate. A cascade ensues through the production of more pairs by γ-rays that originate via the inverse Compton process. The pairs find themselves within the ever-widening magnetic geometry provided by the surrounding disk wind, and this allows the conversion of the catastrophic increase in the relativistic gas pressure into order relativistic bulk motion with Lorentz factors of $\simeq 10$. The resulting motion is not likely to be stationary, and above some threshold the Alfvénic turbulence will impose a knotted character upon the relativistic flow.

The surrounding hydromagnetic disk wind likely plays an important role for the relativistic plasma in providing a *channel* for it. This insulates the pair plasma from the interactions with the surrounding medium. Moreover, the interaction of such a relativistic beam with a thermal plasma of the surrounding flow or ambient medium is strongly quenched as long as the magnetic field in the structure is strong enough. Calculations show that destructive instability under these general conditions sets in at scales ≥ 100 pc (Sol, Pelletier, & Asseo 1989; Pelletier & Sol 1991).

8. CONCLUSIONS

We have shown that disks and winds are intimately connected with one another, since winds are a very efficient means of extracting disk angular momentum. Moreover, wind torques play a dominant role in determining accretion disk structure if magnetic energy densities in Keplerian disks are comparable to their thermal energy and disk surface fields are reasonably open in structure. We have also demonstrated that for the lowest energy state, steady outflows (in which $B_\phi \propto r^{-1}$), there is a unique distribution of magnetic field across the disk. Thus, the structure of outflows and that of their underlying disks are inseparable and should be thought of as comprising a single dynamical system in which both accretion and ejection occur. The gravitational field of the central object is the primary source of energy that is tapped through the accretion process. The magnetic field mediates the process and should be regarded as a "scaffolding" that enables this energy to be mechanically extracted from the accretion flow. Matter falling into this engine ends up having its angular momentum stripped away, and falling into the central object. This is likely to be the means by which star formation proceeds in accretion disks, and may also be central to the physics of massive black hole formation. The basic point is that magnetized gas spiraling into a gravitational potential well readily produce highly collimated, centrifugally driven disk winds.

We have shown that the coupling between disk and wind gives rise to flows in both YSOs and AGNs whose mechanical energy, thrust, mass-loss rate, velocity, and density comply with all known observational constraints (see §§ 6 and 7 for the details). For YSOs in particular, by equating the thrust observed to be associated with large-scale CO flows to that provided by the underlying disk wind, we constrain one of the two free parameters in the problem. We find that the accretion rate and wind mass-loss rate are fundamentally related to one another and have typical values $\dot{M}_w/\dot{M}_a \simeq 10^{-1}$ for both YSOs and AGNs. The magnetic field required for the centrifugal drive is modest in both these systems ($\lesssim 10^4$ G for the inner disk in AGNs and $\simeq 10$ G for the comparable region in YSO models). When the finite pressure in the disk corona is considered, we find that the opening angles (measured from the disk surface) that allow the operation of the centrifugal drive are as much as $\simeq 70°$. We have derived simple formulae from our general theory that are useful for the interpretation of the observations (see § 7 in particular).

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A two-dimensional hydromagnetic disk wind depends upon two parameters, and we have found that these are similar between YSOs and AGNs. Indeed, one of the major points of this paper is the demonstration of the consistency of a non–self-similar disk wind model with these observations. Other major features of these winds include their self-collimation that relies only upon the magnetic pinching forces within the rotating outflow. We have demonstrated that, under a wide range of conditions, flows will moreover recollimate to magnetic focal points (see eq. [5.13]) where shocks may occur and where jets are likely to begin radiating.

We emphasize that the above conclusions hold under the assumption of complete angular momentum extraction by the wind. Within this constraint, they are very robust and are not restricted to some special model for the magnetic field distribution on disks. However, even the inclusion of dissipation due to ambipolar diffusion is unlikely to change these basic conclusions. Calculations carried out for dissipative disks with ambipolar diffusion (Königl 1989) still show that most of the gravitational energy and angular momentum are carried off in the wind. We are currently generalizing this analysis for our own non–self-similar theory. The remaining question about energy extraction is whether or not a disk magnetosphere always remains essentially "open" in its geometry. Finally, accretion from disks should leave central protostars spinning at near breakup values, which is not observed. Nevertheless, stars with magnetospheres can rid themselves of even this angular momentum by coupling with the disk, as proposed by Ghosh & Lamb (1979) and Königl (1991).

In general, we have found that many of the conclusions in the fundamental paper of Blandford & Payne (1982) have counterparts within the much wider classes of non–self-similar models that we have investigated. We have formally added to the theory of disk winds the boundary condition that at every radius upon the disk the wind carries off the disk angular momentum. It is this physics that guarantees the robustness of the results and that fixes the scaling laws of physical quantities throughout the wind and upon the disk. The fact that gravitational energy released in accretion can end up in the form of mechanical outflow rather than heat is a very important constraint on trying to deduce the accretion rates of disks by relying only upon their blackbody emission.

We believe that disk winds are the simplest means of understanding correlations between disk and wind properties now being discovered in YSOs. The implication of this is that the formation of stars, and perhaps even black holes in AGNs, can occur by jet-driven accretion through disks.

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APPENDIX A

DERIVATION OF THE GRAD-SHAFRANOV EQUATION

The right-hand side of equation (2.17) must be parallel to ∇a. Therefore, only the poloidal part of the right-hand side will remain in the equation. The poloidal part of the right-hand side of the equation is thus evaluated:

$$\boldsymbol{u} \times (\nabla \times \boldsymbol{u}) = \boldsymbol{u}_p \times (\nabla \times \boldsymbol{u}_p) + \Omega \nabla(\Omega r^2) .$$

Analyzing the poloidal portion of the Lorentz force in the same way, the right-hand side of equation (2.17) becomes

$$\mathrm{RHS} = \boldsymbol{u}_p \times (\nabla \times \boldsymbol{u}_p) - \left[\frac{\frac{1}{2}\nabla[(rB_\phi)^2] + (\Delta' a)\nabla a}{4\pi\rho r^2}\right] + \Omega\nabla(\Omega r^2) - \nabla(\Omega_* \Omega r^2) , \tag{A1}$$

where $\Delta' \equiv \nabla^2 - (2/r)(\partial/\partial r)$. The last two terms in this equation arise from centrifugal stresses. The leading term is the inertial term due to poloidal motion of the gas and may be written

$$\boldsymbol{u}_p \times (\nabla \times \boldsymbol{u}_p) = \frac{\nabla a}{4\pi r^2}\left[\frac{\rho_A}{\rho^2}\Delta' a + \frac{1}{2}\nabla\left(\frac{\rho_A}{\rho^2}\right)\cdot \nabla a\right] . \tag{A2}$$

Using equation (2.7) for B_ϕ as well as equation (2.9), one finds

$$\frac{\nabla[(rB_\phi)^2]}{4\pi} = (\Omega r^2 - \Omega_* r_A^2)^2 \nabla(\rho_A) + 2\rho_A(\Omega r^2 - \Omega_* r_A^2)[-\nabla(\Omega_* r_A^2) + \nabla(\Omega r^2)] . \tag{A3}$$

This collection of terms is the combined effect of toroidal magnetic pressure and pinching. Gathering all these results together into equation (A1), one finds after some algebra that

$$\begin{aligned} \frac{d\mathscr{E}}{da} &= \frac{1}{4\pi\rho r^2}\left[\frac{\rho_A - \rho}{\rho}\Delta' a + \frac{\rho}{2}\nabla a \cdot \nabla\left(\frac{\rho_A}{\rho^2}\right)\right] - G(a, g; r) , \\ G(a, g; r) &\equiv g\Omega_*\left[\frac{d(\Omega_* r_A^2)}{da} + g\Omega_* r^2 \frac{\rho}{2\rho_A}\frac{d\ln\rho_A}{da}\right] + (1-g)\Omega_* r^2 \frac{d\Omega_*}{da} . \end{aligned} \tag{A4}$$

Equation (2.18) then follows from this equation.

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APPENDIX B

CRITICAL SURFACES

The fast surface are most easily found by first taking the derivative of the Bernoulli equation with respect to the poloidal coordinate s. Differentiating equation (2.16), one obtains

$$\left[\frac{m^2 r_e^4}{r^2}(\nabla\psi)^2 - \frac{\mu g\omega^2}{(m^2-1)^2}\frac{r^2-r_A^2}{r_e^2} + \mu\frac{d\bar{h}}{dm^2}\right]\frac{\partial m^2}{\partial s} = -\mu\frac{\partial\bar{\Phi}}{\partial s} + \frac{\mu\omega^2}{2r_e^2}\frac{\partial r^2}{\partial s}\left[1-\left(\frac{m^2}{m^2-1}\right)^2\left(1-\frac{r_A^4}{r^4}\right)\right] - \frac{m^4}{2}\frac{\partial}{\partial s}\left[\frac{r_e^4(\nabla\psi)^2}{r^2}\right]. \quad \text{(B1)}$$

Using the definition of B_p in terms of the flux function ψ, and the dimensionless density μ, then the dimensionless Alfvén speed in the poloidal magnetic field is

$$\bar{V}_{A,p}^2 = \frac{m^2}{\mu}\frac{r_e^4(\nabla\psi)^2}{r^2}. \quad \text{(B2)}$$

The second term on the left-hand side of equation (B2) simplifies if we recall equations (2.8) and (2.10) and note that the dimensionless Alfvén speed in the toroidal field is

$$\bar{V}_{A,\phi}^2 = \frac{\omega^2 r^2 g^2}{m^2 r_e^2}. \quad \text{(B3)}$$

Substituting equations (B3) and (B2) in the left-hand side of equation (B1), we find that it can be written as

$$(m^4\bar{V}_{A,p}^2 - m^2[\bar{V}_A^2 + \bar{c}_s^2] + \bar{c}_s^2)\frac{1}{m^2(m^2-1)}\frac{\partial m^2}{\partial s},$$

where $\bar{c}_s^2 \equiv m^2(d\bar{h}/dm^2)$ and where the total Alfvén speed in the combined field is $V_A^2 = V_{A,p}^2 + V_{A,\phi}^2$.

The quantity in square brackets appearing in the previous expression can be factored. We first recall that the fast and slow magnetosonic speeds in a magnetized medium are $V_{FM} = u_{p,+}$, $V_{SM} = u_{p,-}$:

$$u_{p,\pm}^2 = \tfrac{1}{2}\{(V_A^2 + c_s^2) \pm [(V_A^2 + c_s^2)^2 - 4c_s^2 V_{A,p}^2]^{1/2}\}, \quad \text{(B4)}$$

from which one derives the useful relation

$$V_{FM}^2 V_{SM}^2 = c_s^2 V_{A,p}^2.$$

By defining

$$n^2 \equiv (u_p/V_{FM})^2, \qquad n_s^2 \equiv (u_p/V_{SM})^2, \quad \text{(B5)}$$

we finally factor the expression and derive the critical surface equation (4.1).

APPENDIX C

ALFVÉN CRITICAL SURFACE

We first multiply the Bernoulli equation by $(m^2-1)^2$ and expand it to first order about the Alfvén surface. Note that we use the full critical point equation (B1) in this analysis. Noting that $m \simeq 1$ near the Alfvén surface and that $m^2 - 1$ is small, one obtains

$$(m^2-1)^2 A = \frac{(r^2-r_A^2)^2}{r_A^4}, \qquad A \equiv \left(1+\epsilon_1 - \frac{c_A}{\lambda}\right), \qquad \epsilon_1 \equiv 2\left[\frac{(\bar{\mathscr{E}}-\bar{\Phi}-\bar{h})r_e^2}{\omega^2 r^2}\right]\Bigg|_A. \quad \text{(C1)}$$

We must now expand the factors on the left- and right-hand sides of the equation about the Alfvén surface. One subtlety in the procedure is that the function r_A must also be expanded about a particular point on the Alfvén surface. The result is

$$(\boldsymbol{r}-\boldsymbol{r}_{A,o})\cdot\nabla m^2|_{A,o}\, A^{1/2} = \pm\frac{2}{r_{A,o}}(\boldsymbol{r}-\boldsymbol{r}_{A,o})\cdot[\hat{\boldsymbol{r}} - (\nabla r)_{A,o}]. \quad \text{(C2)}$$

Equation (4.2) in the text follows immediately from this.

The Grad-Shafranov equation can be analyzed in a similar fashion. Near the Alfvén surface it reduces to

$$(m^2-1)(c_2 - c_3\lambda) = 2\lambda\frac{(r^2-r_A^2)}{r_A^2},$$

$$c_2 = r_A(\nabla m^2\cdot\nabla r_A)|_A = r_A|\nabla m^2|_A c_A^{1/2}\cos\theta_A \equiv C_2\cos\theta_A, \quad \text{(C3)}$$

$$c_3 = \frac{n_1}{2n_o}\left[1+\epsilon_2\left(\frac{r_A^2}{r^2}\right)\right],$$

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where ϵ_2 is defined in equation (2.24) and where θ_A is the angle between the field line and the Alfvén surface. Equation (4.3) in the text follows immediately from this result.

APPENDIX D

THE SM SURFACE

The regularity of the solution at the point $n_s = 1$ is found from the critical point (eq. [4.1]). In the limit $m^2(r_A/r)^2 \ll 1$,

$$-\left.\frac{\partial \bar{\phi}_{\rm eff}}{\partial s}\right|_s - \left.\frac{m_s^4}{2\mu}\frac{\partial b_p^2}{\partial s}\right|_s = 0 , \tag{D1}$$

where

$$b_p^2 \equiv B_p^2/(a_e^2/r_e^4)$$

is the dimensionless poloidal field and where we have used the relation

$$\bar{V}_{A,p}^2 = b_p^2 m^2/\mu . \tag{D2}$$

The Alfvén Mach number at the SM point may be derived by noting the scaling

$$n_s^2 \simeq (u_p/c_s)^2 = m^2(V_{A,p}/c_s)^2 . \tag{D3}$$

Combining this with equation (D2) then gives

$$n_s^2 = \frac{m^4 b_p^2}{\mu \bar{c}_s^2} . \tag{D4}$$

The condition that $n_s = 1$ at the sonic point then immediately gives equation (4.12) in the text.

The position of the sonic point along a field line can be found using equation (D1). Expanding the effective potential to first order (as discussed in § 4.3), we find a differential equation satisfied at the SM point. It is useful to use dimensionless variables scaled with the disk half-thickness $z_o = c_s/\Omega_o$. Thus, with $y \equiv z_s/z_o$ and $x \equiv \delta r_s/z_o$, one finds that at the SM point equation (D1) gives

$$y = 3x\frac{dx}{dy} - \frac{\partial \ln b_p^2}{\partial y} . \tag{D5}$$

This equation is of the same general form found by BP (see their eq. [4.6]).

In order to solve for the position of the SM point, it is necessary to have the shape of the field line. The GS equation provides this. It is easy to show that it is essentially force-free near the disk. Knowing $y(x)$ near the disk from this equation, the analysis can be completed. Lacking a numerical solution for ψ, we defer this exercise to a forthcoming numerical paper.

APPENDIX E

MORE ON RECOLLIMATION

Consider case 2 of § 5.2 ($\lambda/\chi^2 \gg 1$). Then χ_t satisfies

$$\eta(\chi_t)\lambda + [\eta(\chi_t)\lambda]^{1/2}\kappa_r = \frac{2\lambda}{n_o^2}\frac{\chi - 1}{\chi^3} . \tag{E1}$$

Since $\eta < 0$ for sufficiently large χ and $\alpha < 3$, this regime has more restricted applicability. The solution for sufficiently large χ and $\alpha > 3$ is

$$n_t^2 \simeq \frac{2^{3/2}\lambda^{1/2}}{[(\alpha/3) - 1]^{1/2}\kappa_r} - 2 , \tag{E2}$$

a result which is again similar to that of BP. Note in particular that the FM solution that is characterized by a constant $\chi = 3/2$ cannot recollimate, which explains the result of § 5.1.

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